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Faculty of Sciences

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Mathematical models of T cell proliferation, with potential applications to data

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DECLARATION

This work was carried out at the University of Yaounde 1 in partial fulfillment of the requirements for a Master of Science Degree.

I hereby declare that except where due acknowledgment is made, this work has never been presented wholly or in part for the award of a degree at the University of Yaounde 1 or any other University.

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DEDICATION

To my family and especially to Dr. Fernando Kemta Lekpa

List of abbreviations

ARE .... Average relative error

CFDA,SE .... Carboxyfluorescein succinimidyl ester (nonfluorescing)

CFSE .... Carboxyfluorescein succinimidyl ester

CTL .... Cytotoxic T lymphocytes

DDE .... Delay differential equations

DNA .... Deoxyribonucleic acid

DMSO .... Dimethyl sulfoxide

FCS .... Fetal calf serum

HSCs .... Hematopoietic stem cells

IBS .... Individual based simulation

CKE .... Chapman Kolmogrov Equation

MHC .... Major histocompatibility complex

NK .... Natural killer

ODE .... Ordinary differential equation

PBS .... Phosphate buffered saline

RMPI .... Roswell Park Memorial Institute

SSD .... Sum of square of differences

SME .... Stochastic Master Equation

TCD4 .... Helper T-cells

TCD8 or CTL .... Killer T-cells

Abstract

It is well known the fundamental importance T cells in the immune response. A good immune response depends not only qualitatively, but also quantitatively of the number of T cells produced after been assaulted by a foreign invader. Therefore it becomes necessary to understand the process underlying T cells proliferation. In this work, we review biological insights of T lymphocytes division such as the fact the activation process move the cell from the quiescent stage to the proliferation stage. We further present the earlier mathematical models of T cells dynamics and introduce a new model based on the stochastic nature of the T cells dynamic. We only consider the case where the activation parameter is generation dependent. We end by deriving some statistic such as the means and the variances of cell counts and set up the scene for a fitting to real data.

Resumé

Il est bien connu à quel point les lymphocytes T sont importants pour la réponse immunitaire. L'efficacité d'une réponse immunitaire spécifique dépend non seulement qualitativement, mais aussi quantitativement du nombre de lymphocytes T produits après avoir rencontré un antigène. De ce fait, il devient capital de comprendre le processus de prolifération des lymphocytes T. dans ce travail, nous faisons revue sur la biologie des lymphocytes, tout en presentant le procédé expérimental utiliser pour tracker leur division invitro et invivo. Ensuite, nous nous apaisons sur le processus d'activation qui conduit la cellule du stade latent au stade de prolifération. Nous présentons en outre quelques modèles mathématiques antérieurement utilisent pour décrire la dynamique des lymphocytes T. Sur la base des limitations des précédents modèles, nous proposons un nouveau modèle basé sur la nature stochastique du processus de prolifération. Nous ne considérons que le cas où le paramètre d'activation dépend de la génération et non du temps. Pour finir, nous extrayons certains paramètres statistiques tels que les moyennes et les variances du nombre de cellules tout en préparant le cadre pour un ajustement du modèle aux données réelles.

Contents

1. Introduction

Throughout history humankind works to ameliorate their living conditions and in particular their sanitary well-being . Through the development of science and technology, many steps toward better conditions had been made and in the special case of biology, Immunology field has a long history of discoveries that lead to the eradication of many diseases such as smallpox. Before we go further let us pause for a few remarks from the history of immunology. Then fist records of the immunology started in 430 BC in Athens where about 25 % of the population was dying of plague until it was noticed those recovering from the disease was not dying from a second exposure. Later the first vaccine was due to experiment by Edward Jenner (1749-1843), who infected a child with cowpox (cow version of smallpox which is less damaging for human as smallpox) and then later infected the same child with smallpox and the child survived. This discover inspired Louis Pasteur(1822-1895) who came out with **germ theory**. The theory is based on the following postulats

- Each disease is caused by different types of pathogen
- The immunity is specific to the types of pathogen.

Base on the above ideas, Pasteur developed attenuated form of pathogen to stimulate the immunity against the identified pathogen. With this approach, vaccines for many diseases were able to be designed. Among other we have smallpox, diphteria, tuberculosis, tetanus,yellow fever, polio and more recently in 2006 HPV vaccine. Even though the germ theory and immunity seems to work, we still lack some knowledge on how does immunity work.

From the above observation, we can conclude that there are still many challenges that the solutions could lie in the better understanding of the immune system. To make it more concrete and to understand why it is so important more than ever to put an accent in immunology, let us give some figures for the only case of HIV AIDS. According to UNAIDS data for the year 2017, about 36.9 million people were living with HIV and about 1.8 million people became newly infected and 940,000 died from AIDS-related illnesses. In addition, we can charge US $ 26 billion was spent for HIV and related programs. Immediately, the question arisen is what about other diseases and their cost for humanity? Fortunately, there is hope because there are more people in the field of immunology, additionally recent developments in other disciplines such as mathematics and computer sciences provide more possibilities beyond experimental work. Furthermore, the codisciplinary approach and funding are working than ever. Therefore the contributions are needed to overcome the challenge of diseases, cancers and other auto-immune diseases.

While some are working towards the development of strategies to stop the spread of diseases by focusing on the transmission and vectors, others are working on the production and amelioration of medicines and vaccines. The second choice passes through the better understanding of the immune system which can be seen as a big puzzle or labyrinth in where nobody knows where to go. But as we said the scene is set up to come out with some breakthroughs, which passe through the development of new theories. Sometimes, the theory could appear to be wrong and apply for many years. For an instance, we have the case of the Linus Carl theory of antibodies which was found to be wrong after two decades of use. To avoid such a situation, the theory must be put in the testable form which we think the best way is mathematical model. Furthermore, mathematical modeling allows us to derive new knowledge which is impossible to derive experimentally. In the framework of the understanding of the immune system, it is important to focus on one of the major components of the cells which intervene almost in every aspect and components of the immunity; it is notably T lymphocytes also known as T cells. Studying T cells' dynamic is part is the fundamental questions:

- What is immune system?
- How it works?

The goal of this essay is not to fully answer these questions, but rather to give a contribution to that could help to answer such questions. Given that the immune system is a big machine and part has its own full description, we only focus on T cells, because of the major work they play within the machine. We investigate the quantitative behavior of T cells in response to an attack by starting with a brief description of immune system and immunity inspired by [1] and [2]. We provides in the framework of existing knowledge of cell division and immunology [3] a protocol for T cell labeling (CFSE labeling). Based on the insight from CFSE labeling, we propose a mathematical model of T cell dynamics based on stochastic master equations (SME), we make use of the probability generating function (PGF) to extract the mean and the variance of cell counts. We end by constructing a framework for maximum likelihood estimation approach to fit the obtained model to data. A global picture of this work is described below (see Figure (1.1))

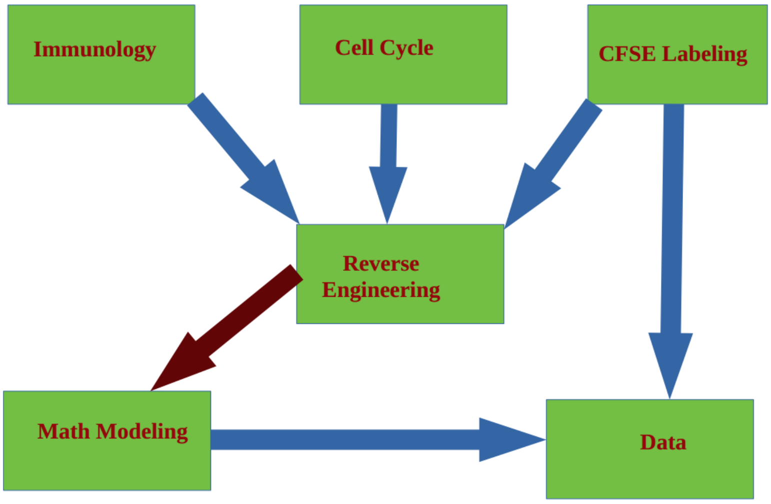

Figure 1.1: Global description of the modeling approach

2. Background and motivation

2.1 Immune system

The Immune system can be defined as the network of cells, tissues, and organs that work together to defend the body against attacks by foreign invaders [1]. The Immune system is very powerful and one of the best ways to think about it is to imagine how many bacterias, fungi, virus and other threats that we encounter every day. If regardless of those attacks, we are not ill, it is because the immune system protects us by providing a range of response against such aggression. However, there are times where its protection becomes inefficient and that leads us to a stage of illness and in the worst case to death. To avoid such a situation, we need through medicines and vaccines help our immune system to give efficient immune responses. The questions that one may ask is how to develop good medicines and vaccines? The answer to this questions lies in the better understanding of the immune system. The fully functional immune system involves so many organs, molecules, cells, and pathways in such an interconnected and sometimes complicated process that it is often difficult to know where to start[2]. This is why it will be too pretentious to say we can completely describe it. However, the recent advancements in science and technology allow us to have better knowledge and description of the immune system. In the framework of this work, will give a simple description that allows us to understand what we are talking about and that fit the context of this work. Before going into the description of the immune system, let us quote here why is so important to understand the immune system while understanding diseases transmissions, genetics, human metabolism, and pathogens behaviors. We will make our point by reminding you of a historical fact: the eradication of smallpox. Smallpox was a contagious viral disease. It was one of the most devastating diseases in the world. According to the WHO, during the twentieth century, it was estimated that smallpox was responsible for more than 300 million deaths. The eradication was made possible because people focused on the understanding of the immune system and the structure of the virus. Which allowed to develop an early version of vaccine and then later an improvement of the effectiveness of the vaccine which leads in 1980 to the complete eradication. From this example, it is clear that it is important to also develop immunology knowledge if we want to tackle the challenge of diseases.

A simple picture of the immune system consists to divide it into two branches namely the **innate immune system** and the **adaptive immune system**. Before we go further, it is important to point out that immunity turns around the notion of **self** and **nonself**. The self refers to particles, such as proteins, molecules, and even cells, that are a part of our body or made by our body. and nonself is foreign cells (cells that do not carry the self-makers). The innate immune system is in forefront of the immune system, and it is made of defenses that can be launched immediately once we assaulted by a pathogen. It is essentially made up physical barriers such as skin, liquid defense mechanisms such as mucous and sweat. We can also add the general immune response which is characterized by the inflammation, complement and nonspecific cellular response through cells of the innate immune system. Those cells mainly come from the myeloid lineages of hematopoietic stem cells (HSCs), among them we can mention phagocytic cells such as macrophages and neutrophils. Their role is to give a quick response by swallowing the "foreigner" and digest it (phagocytosis). Another types of cell in this branch are mast cells which mediate phenomena such as hypersensitivity and allergic reactions. One of the important component are granulocyte cells such as eosinophils, basophils that spray chemicals to destroy the encounter threat. The Natural Killer cells (NK) are also an important type of the innate branch; they are armed with granules filled with chemicals that are used to destroy the targeted invaders. The clear limit between the cells of the innate immune system and the adaptive immune system is difficult to draw as some of the cells such as **dendritic cells** known as belonging to the innate branch are responsible for the initiation of adaptive immune responses. The last part of the innate immunity is known as the **complement**, made of proteins. They bind common pathogen-associated structures and initiates a cascade of labeling and destruction events [2]. As mentioned before, the innate immune system does not give a specific response to a threat and the given rapid response to be put in place sometimes can eliminate infectious agents within hours of encounter [2]. It always launches the same immune responses regardless of the type of pathogen: this is why it is sometimes called the nonspecific immune system. Although the innate immune system is powerful, it is not always victorious in its battles; so the adaptive branch must take over, especially when it is a previous encounter antigen.

The adaptive immune system is activated when the innate immune response is unable to contain an infection. Indeed, without information from the innate branch, the adaptive response could not be mobilized. This type of immunity is much slower than the precedent; the immune response is given against the pathogen within 5 or 6 days after breaking the first barrier [2]. This branch gives a very specific response to the encountered threat, powerful and especially generates a memory for a faster response to the future attacks by the same pathogen. At the first encounter with the threat, adaptive immunity provides a primary response, during which the appropriate lymphocytes will be clonally selected, and used to eradicate the pathogen while some of them are kept as memory cells for the future attacks by the same pathogen. During the subsequent response, the memory cells are re-enlisted to fight again while learning and improving the fighting strategy for the next attacks. Given that in this essay, we are interested in T cells, which belong to the adaptive arm, let us give a further description of this branch.

As mentioned above, sometimes, for the effective success of immune response, the adaptive arm must take over. This happens when a signal is sent by the innate branch generally in the form of molecules more known as cytokines.

The adaptive immunity is also known as acquired immunity lies on two types of cells namely B cells and T cells. These cells allow us to divide the acquired immune system into two sub-branches: the **humoral immunity** directed by B-cells and the **cell-mediated immunity** controlled by T cells. The main role of B cells is to secrete proteins called **antibodies**. Antibodies have mainly three functions to play: firstly, by a process called neutralization they bind to the pathogens to reduce their infectivity. Secondly, they bind to the threats and facilitate it phagocytosis: this is commonly known as **opsonization** and lastly, they help the complement to achieve its cascades of destruction. Since B cell is the main actor within the humoral immunity, let us give further details that will allow us to better understand their role. B cells take its origin in the bone marrow from a common lymphoid progenitor descendant of HSCs and remain in the bone marrow until the differentiation process starts. Once the differentiation process has begun, the immature B cells leave the bone marrow and migrate towards the spleen where they will complete their differentiation. Differentiation allows them to express their immunoglobulin receptors which allow them to recognize a specific type of antigen. Once the antigen is recognized, the B cells secrete the corresponding antibodies which by their complementary structures will either bind to the antigen to form the antigen-antibody complex or will inhibit the action of the antigen which will be more easily phagocytized. For all of this to happen, the immune system has to undergo several processes that involve several molecules and cells such as T cells (for further description of B cell's mechanism, see [2]).

T cells is the central core of the acquired immunity. In fact they orchestrate the functions of the other cells and molecules involving within the defense process. For example, the interaction between a naive T cell and an antigen-presenting cell is the initiating event of the adaptive immune response [2]. Before showing how that is done, let us first understand their genesis. Both T and B lymphocytes come from the lymphoid branch of HSCs known as lymphoid precursor. The decision of a lymphoid precursor to give rise to a T cell depends on a receptor called **Notch**. Their development processes in the bone marrow are similar to that of B lymphocytes, with the only difference that their maturation and differentiation takes place in the thymus. In the thymus, T cell development begins with a small numbers of lymphoid precursors migrating from the bone marrow and blood into the thymus, where they will proliferate, differentiate, and undergo selection processes that result in the development of mature T cells. T cells contribute to the immune response in two major ways which will allow us to classify them into two categories. The first categories known as helper T cells (TCD4) which direct and regulate immune responses by communicating with the other types of cells. For an instance, some TCD4s assist nearby B cells in the production of antibodies, while some contribute to phagocytosis. The second categories are Killer T cells also called cytotoxic T lymphocytes (CTLs or TCD8) directly attack infected or cancerous cells which lead them to apoptosis. This second types of T cells are reputed for their intervention against viral infection since they can recognize the infected cell. Generally, T cells only recognize an antigen if there is an expression of major histocompatibility complex (MHC) proteins on the cell surface. MHC molecules are proteins recognized by T cells when distinguishing between self and non-self. Some "foreigners" antigens are useful for our well-being and they do not carry the self-makers and should be fought. However, this does not happen because there is a particular sub-type of T cells called regulatory T cells also known as suppressor T cells that regulate the immune response and enable these useful antigens to survive within the body cells.

From the foregoing, it is clear that T-cells are important for well-being because they intervene almost in all stages of the immune response. In order to properly fill their mission, it is important for them to proliferate when they encounter the appropriate stimulus, in the form of antigens or cytokines. In some cases, it is necessary to artificially enhance T-cell proliferation (e.g. by using drugs) to enable a more effective response. Therefore, it is important to understand the mechanism behind their proliferation. The earlier approach was to use experimentation to understand their those mechanisms, but it just tells us little about the processes. This is why we need modeling to make sense of data produce by experimentation. A number of experimental techniques and models have been successfully developed and applied to investigate the mechanisms of T cell proliferation. These techniques and models will be review below.

2.2 Experimental analysis of T cell proliferation

How cells proliferate is still not well understood, in the earlier age of biology, we knew that the proliferation is guided by cell division according to the exponential division. The development of experimental methods has revealed that the division process is more complex. For example, it is not every cell that is suggested divide, some will remains quiescent along the division process while some will directly undergo apoptosis. Even though it still not well understood, the recent developments of the experimental biology lead to some advancements. A number of techniques have been developed so far to track cell division in vivo and in vitro. One of the common one use especially in vivo is to quantify the tritiated thymine found in DNA of dividing cells. This technique just accounts for the number of division, but lack to tell us information about the history of the cell [4]. Checking the reduction of the 3-(4,5-dimethylthiaxol-2-yl)-2,5-diphenyl tetrazolium (MMT) bromide is also well known for quantification of cell division, but the problem is that the MMT is sensitive to the activation state of the cells. The 5-bromodeoxyuridine is also famous for cell counting, but it is unable to make a difference between cells' generations. As these methods tell us little about what we want to know, the carboxyfluorescein diacetate succinimidyl ester (CFSE) dye has is becoming the most used technique to understand cell division. The use of CFSE labeling techniques to understand lymphocyte dynamics has become popular after the work of Weston and Parish in 1990, where they used it to describe the migration of lymphocytes in vivo. It is for this reason that certain pioneers of the experimental field such as Gett and Hodgkin [5] use it to show that the number of division undergone by the cells after activation is a critical element in T cell differentiation, then in 2000 the same authors use it to investigate the time taken for the first division and the subsequent divisions. Meantime, many theoretical advances are based on the analysis of data from CFSE labeling. Among these, we can list the cascade of models due by Rob J, de Boer, Alan s. Perelson and Vitaly V. Ganusov ([6],[7], [8], [9],[10],[11], [12]) who tried to give a sense to the experimental data. In this part, we propose to revisit and discuss some of these experimental results based on CFSE labeling that the experimental protocol is given in appendix

2.2.1 Experimental results on T cell Proliferation

For nearly 3 decades, one of the biggest challenges in experimental biology is that of understanding how cells divide in general and more particularly the case of T cell. The investigations on how T cells and B cells divide is, in particular, has been of paramount interest. In order to answer this question, many advances have been made. Especially since it has been revealed from the work by Weston and Parish [13] that the CFSE labeling technique can be used to monitor Immune cells division. Many pioneers in the field of cell biology and immunology have provided interesting insights into this issue. We are reviewing and discussing some of these insights here.

Under normal conditions, T cells are in the quiescent stage (no division), when they receive an external signal, they activate and divide. the process that governs division after activation is the same as that of other cells. For any eukaryotic cell to divide, it must go through two major events defining the cell cycle: **mitosis (M phase)** and **interphase**. During each of this phase, a cell performs particular tasks. In the interphase, the cell grows and makes a copy of its DNA, while during the mitosis, the cell separates its DNA into two sets and divides its cytoplasm to form two new cells. our main interest here is to focus on the complex part of cell cycle namely interphase. The interphase takes place between two mitoses, which justifies its name. This event is subdivided into 3 phases. The first one G_0/G_1 also known as first gap phase follows directly the previous mitosis and is characterized by cell growth, the copies of organelles are made, and the preparation of the materials for phase S. The second one is the S phase known as the phase where occurs the synthesis of the DNA. During this phase, the cell synthesizes a complete copy of the DNA in its nucleus, it also duplicates the microtubules' organizing structure called centrosome that will help separate the DNA during mitosis. The last phase is called the G_2 phase or second gap phase, it is characterized by a secondary growth of the cell and the verification of the genetic material before the division which will be done at the next stage of the cell cycle which is the M phase. Regarding the M phase, it is composed of two important points: the nucleus division called mitosis and the cytoplasmic division called cytokinesis. It is also important to notice that G_0/G_1 are also called checkpoints because it is in this stage that the DNA materials are checked.

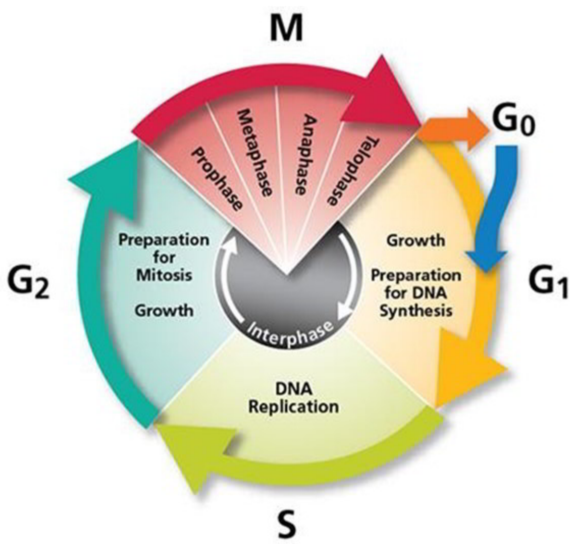

Figure 2.1: Description of different phases of cell cycle and events occuring (Image from https://www.earthslab.com/physiology/the-cell-cycle/

As mentioned above, our first interest is what is happening in the interphase, Stern and Nurse [15] addressed this question by focusing on the S phase. By investigating fission yeast cell cycle, they found out that major determinant of the cell cycle is a protein kinase known as $cdc2$. Indeed, $cdc2$ in combination with another protein called $cdc13$ determine whether the cell will enter mitosis. This happens by controlling the onset of S phase and also guarantee that there is only one S phase by cell cycle. This happens by the level of $cdc2$ at the different stages of cell cycle. From that they where two hypothesis made. The first hypothesis was based on the fact that there are two unique molecules corresponding to the passage from G_1 to S and from G_2 to M. The second hypothesis is based on the rate of phosphorylation dissociation constant. This second view seems to be more reasoning. Let us remind that for DNA to bind to the promoter, it should be no phosphorylation, whereas for transcription to take place, the phosphorylation is important; and the $cdc2$ protein is responsible for the phosphorylation.

Definition 1. Rate of phosphorylation dissociation constant is defined as the threshold of $cdc2$ concentration that enable half of the available protein to be phosphorylated

The amount P of the protein to be phosphorylated is given by the following formula

$$P = \frac{C^{st} \cdot C}{C^{st} \cdot C + km} \tag{2.2.1}$$

Where Km is rate of phosphorylation dissociation constant, C is the concentration of $cdc2$ and C^{st} is an arbitrary constant. Let P_1 and P_2 been respectively the thresholds concentration of $cdc2$ needed for the passage to the S phase and to the M phase, Km_1 and Km_2 their corresponding rate of phosphorylation respectively. According to the second theory stated by Stern and Nurse [15], if $Km_2 > Km_1$ then the concentration of $cdc2$ needed to phosphorylate P_2 is larger than the concentration needed to phosphorylate P_1 then the cell will enter in Mitosis. To summarize this theory, a low amount of $cdc2$ concentration ends the mitosis follow by a moderate amount which initiates the transition from G_1 to S. This amount is maintain below a certain threshold, that guarantees the good evolution of the S phase and blocks the possible return to G_1. A subsequent increase of $cdc2$ concentration leads the cell to mitosis.

Another interesting result about T cell dynamic, lies on the work by Gett and Hodgkin [16]. In this work based on quantitative division-tracking methods using CFSE dye, it was found that T cell divide according to asynchronous division progression characterize by different peaks of CFSE intensities. The two potential source of variation was the division rate and the time to the first division which can be explained by heterogeinety. A developed model called Three-parameter model of T cell activation showed that T cell take time complete the first division after what the subsequent division are crossed approximatively in a constant rate. However the response to stimulation by the cells was not uniform, but the variability to complete the first division can be represented by normal distribution. Therefore the proportion of cells in each division stage at any sampling time can be represented by a model that requires only three variables namely the average time for the first division and his corresponding variance and the average time for the subsequent divisions. These parameter was extracted from time series data of CFSE labeling and incorporated to the model. It resulted that the model was in perfect agreement with conducted experiment. The parameters behavior on change of stimulation condition was also investigated and it appears that on costimulation conditions, the time to the first division was reduced under certain stimuli for example anti-CD3. The same study have shown that the activation toward the first division and dying before division were independent. As a conclusion, The average time taken to cross the first division depends of the stimulation strength receive through T cell receptors and the time taken for the remaining divisions is influenced by the concentration of the interleukine 2 (IL-2). Those conclusions suggest that *two signal theory of lymphocyte activation* by Lafferty and Cunningham [17] must be revisited. However very recent advances have been made in the field and brought up interesting results that deserve to be looked at.

Following the work by Heinzel et al. [18] which have been discussed by Asquith and de Boer [19] and updated by Heinzel et al. [20], we have a better overview on T cell proliferation. Before those mentioned works, it was already known from the study made by Marchingo et al. [21] that stimulated T cells count how many divisions they undergo before returning to the quiescent state. But from the work by Heinzel et al. [18], it has been revealed that the number of division that T cells undergo before becoming quiescent is directed by the expression of the Myc proto-oncogene. And they showed that at the time Myc proteins levels drop below a critical threshold, T cells stop dividing even though the stimuli is removed. The explanation of that is based on the following hypothesis: A sufficient quantity of Myc proteins are produced after the activation and will decrease after each division until a certain threshold is crossed and division stops. Nevertheless, the level and timing of Myc proteins produce after the stimulation depends on the strength and the combination of the signal received. That just raises a question on the perfect understanding of the division process under Myc protein control. To address this question, Asquith and de Boer [19] suggested two possible models. The first each cell undergoes a fixed number of division before Myc proteins drop below threshold. The second based on the fact that Myc proteins declines with time independently of the number of division until the threshold is crossed. The second model seems to be more valuable given that Cyton model proposed by Wellard et al. [22] is based on the same philosophy with the first model. Furthermore, the recent work by Heinzel et al. [18] suggests that is time related. Therefore, T cell proliferation is directed by Myc proteins which the amount decreases over the time at a predicable rate that the subsequent generation will inherit Heinzel et al. [20]. How to take into account the fact that cell might go to apoptosis during the division process even though the *Myc* proteins level is above the threshold? Heinzel et al. [18] suggested some direction. For example she suggests that an analysis by taking into account the heterogeneity might bring up interesting insights. All these questions raised by the experimental approach that leads to hypotheses and theories must be put on the testable and analyzable form. Thus, it appears that the best way to do it is by putting our assumptions in the form of mathematical models.

2.3 Mathematical models of T cell proliferation

One of the popular approaches to understand or extracting knowledge from complex phenomena such as biological processes, ecological interactions and many other more complex systems is the use of mathematical modeling. This has been the case for the understanding of the T cells proliferation dynamics. Many mathematical models have been developed to address this question, each in its own time, and with available biological knowledge. Some people such as, Edwin D. Hawkins, Phillip D. Hodgkin, Rob J de Boer, Alan Perelson, Vitaly V. Ganusov, have made the modeling of lymphocyte dynamics a flourishing field of knowledge that deserves special interest. In this work, we revisit some of these mathematical models. These include Linear Birth Death ODE model, Smith Martin model, Cyton model, Age dependent branching process and more recently an inspiring model from the master's thesis of Juwara [23]. We will review these models base on my previous work (see Foko [24]).

2.3.1 Smith-Martin like model

The Smith-Martin model was used for the first time by J. A. Smith and L. Martin to model the cellular division. This approach essentially takes into account interphase and mitosis, except that instead of dividing the cell cycle into mitosis and interphase, the authors propose a two-phase division: the A phase which is made up of G_1 or else a portion of G_1 (indeterminate length) and the B phase which is S, G_2, M and sometimes a portion of G_1 which has a well known duration (deterministic). According to the model, for a cell to divide, it must move from the A to the B. Many modifications have been made to the original model to have the one that we are going to discuss here. Among these modifications, we can mention the one of De Boer and Perelson [6], De Boer et al. [7], Lee and Perelson [25], De Boer and Perelson [8]. The model that we study here is based on the compartment model.

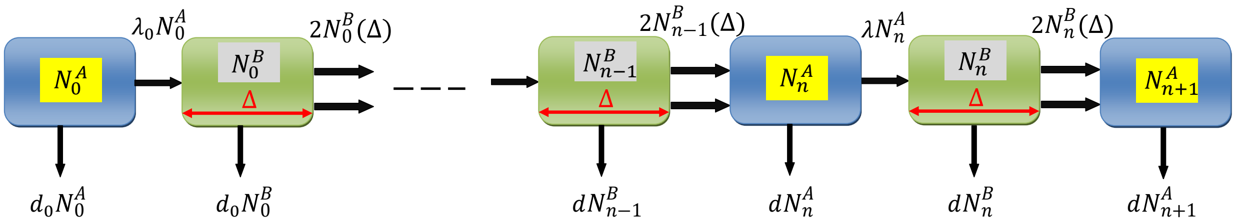

Figure 2.2: Compartment description of Smith Martin model [24]

Let $N_n^A(t)$ and $N_n^B(t)$ represent respectively the number of cells present in A phase and B phase in the nth generation at time t, λ_0 and λ the rates of cells leaving the A phase to the B phase corresponding for the 0th generation and all the generations n $(n > 0)$. To take into account the death, we denote by d_0 and d the death rates for the 0th generation and all the generations n $(n > 0)$ and Δ de duration spend by the cell in the B phase. By following the technique of compartment modeling, we end u with

$$\begin{cases} \dot{N}_0^A(t) = -(\lambda_0 + d_0)N_0^A(t) \\ \dot{N}_0^B(t) = \lambda_0 N_0^A(t) - \lambda_0 e^{-d_0\Delta} N_0^A(t-\Delta) - d_0 N_0^B(t) \\ \dot{N}_1^A(t) = 2\lambda_0 e^{-d_0\Delta} N_0^A(t-\Delta) - (\lambda + d)N_1^A(t) \\ \dot{N}_n^A(t) = 2\lambda e^{-d\Delta} N_{n-1}^A(t-\Delta) - (\lambda + d)N_n^A(t), \quad n \geq 2 \\ \dot{N}_n^B(t) = \lambda N_n^A(t) - \lambda e^{-d\Delta} N_n^A(t-\Delta) - dN_n^B(t), \quad n \geq 1. \end{cases} \tag{2.3.1}$$

Given that it isquite impossible to distinguish the A phase to the B phase, and by only focus on the number of cells present in each generation at a given time, we have

$$\begin{cases} \dot{N}_0^A(t) = -(\lambda_0 + d_0)N_0^A(t) \\ \dot{N}_0(t) = -\lambda_0 e^{-d_0\Delta} N_0^A(t-\Delta) - d_0 N_0(t) \\ \dot{N}_1^A(t) = 2\lambda_0 e^{-d_0\Delta} N_0^A(t-\Delta) - (\lambda + d)N_1^A(t) \\ \dot{N}_1(t) = 2\lambda_0 e^{-d_0\Delta} N_0^A(t-\Delta) - \lambda e^{-d\Delta} N_1^A(t-\Delta) - dN_1(t) \\ \dot{N}_n^A(t) = 2\lambda e^{-d\Delta} N_{n-1}^A(t-\Delta) - (\lambda + d)N_n^A(t), \quad n \geq 2 \\ \dot{N}_n(t) = 2\lambda e^{-d\Delta} N_{n-1}^A(t-\Delta) - \lambda e^{-d\Delta} N_n^A(t-\Delta) - dN_n(t), \quad n \geq 2. \end{cases}, \tag{2.3.2}$$

where

$$N_n(t) \quad = \quad N_n^A(t) + N_n^B(t), \quad n = 0, 1, 2, 3, \cdots \tag{2.3.3}$$

Here Δ can be seen as the delay, but for slowly dividing cell the duration of the A phase can be neglected compared to the full length of cell cycle [6] then the proposed Smith-Martin like (2.3.2) model acquires the form of Linear Birth-Death ODE model

$$\begin{cases} \dot{N}_0(t) = -(\lambda_0 + d_0)N_0(t) \\ \dot{N}_1(t) = 2\lambda_0 N_0(t) - (\lambda + d)N_1(t) \\ \dot{N}_n(t) = 2\lambda N_{n-1}(t) - (\lambda + d)N_n(t), \quad n \geq 2. \end{cases} \tag{2.3.4}$$

The problem with such model is that they fail to distinguish between the A phase and the B phase which can be done experimentally [6].Furthermore, the model fails to capture the death process which cannot be neglected. As a results Hawkins et al. [26] propose a new model that could correct such a limitation.

2.3.2 Cyton Model

The Cyton model is the result of a succession of stages of experience and modeling by a team leads by Philip Hodgkin. They propose a simple model and easier to implement, where the waiting time to first division follows the normal distribution. But the later version known as Cyton model developed by [26] can be adapted to any distribution. To be Let able to build the model, we need to recall the assumptions on which the model is built:

(i) division and death are random/stochastic processes each associated with a specific continuous distribution (probability density function);

(ii) both processes are statistically independent but compete to determine the fate of the cell;

(iii) division events occurring one after another reset the mechanism responsible for these processes;

(iv) only a given fraction of cells after each division are capable to duplicate into the next generation;

(v) the mechanisms are division dependent (For further discussion of this assumptions, refer to [22]).

Let us denote by $n = 1, 2, 3, \cdots$ the number of the division undergone by the cells. To characterize the time to divide and the time to die corresponding to the nth generation we represent them by the continuous distribution function $\phi_n(t)$, and $\psi_n(t)$ respectively. Thus, the probability for a cell to survive at time t when it enters the nth division is

$$1 - \int_0^t \psi_n(\tau) d\tau. \tag{2.3.5}$$

Similarly, the probability that no cell division occur at time t, given that cells are entering the nth division is

$$1 - \int_0^t \phi_n(\tau) d\tau. \tag{2.3.6}$$

However, conducted experiments have shown that lymphocytes do not respond in a similar way to the stimuli. Therefore [26] defines the progression factor f_n which represents the probability of a cell to divide in response to the excitation. Thus, the probability for a cell to survive at time t once enter in the nth division is given by

$$f_n \cdot \left(1 - \int_0^t \psi_n(\tau) d\tau\right), \tag{2.3.7}$$

and the probability that no cell division occurs at time t as cells are entering the nth division becomes

$$1 - f_n \cdot \int_0^t \phi_n(\tau) d\tau. \tag{2.3.8}$$

If we denotes by $N(0)$ the numbers of cells of age zero at time $t = 0$. The rate of cell division in the first step at time t is equal to $N(0)$ times the probability of not dying up to time t times the probability of dividing at time t $(f_0 \cdot \phi_0)$ times the total number of cells.

$$r_0^{div}(t) = f_0 \cdot N(0) \cdot \left(1 - \int_0^t \psi_0(\tau) d\tau\right) \cdot \phi_0(t), \tag{2.3.9}$$

the rate of cell death in the first division at time t is equal to $N(0)$ times the probability for not being divided in the stage zero times the probability of dying after age stage zero

$$r_0^{die}(t) = N(0) \cdot \left(1 - f_0 \int_0^t \phi_0(\tau) d\tau\right) \cdot \psi_0(t). \tag{2.3.10}$$

By taking into account that after a division the rate of cells is doubled, the rate of cells entering the nth division that are subjected to divide is given by $2r_{n-1}^{div}$. Accordingly, parts of a corresponding population divide or die with a certain probability, i.e. one may consider cells entering the nth division at time t' as a set of cells split into two parts: one that divides and another that dies. Therefore, the rate of cells entering the nth division within an infinitesimal time interval $d\tau$ is equivalent to the rate $2r_{n-1}^{div}$ of cells entering the $(n-1)$ division multiplied by the probability that cells survive after the $(n-1)$ division which last $t - \tau$, multiplied by the probability of dividing at time $t - \tau$. Thus, the total rate r_n^{div} at time t corresponds to the integral over intermediate rates

$$r_n^{div}(t) = 2f_n \cdot \int_0^t r_{n-1}^{div}(\tau) \cdot \left(1 - \int_0^{t-\tau} \psi_n(\zeta) d\zeta\right) \cdot \phi_n(t - \tau) d\tau. \tag{2.3.11}$$

Similarly, for dying cells, one has

$$r_n^{die}(t) = 2 \cdot \int_0^t r_{n-1}^{div}(\tau) \cdot \left(1 - f_n \int_0^{t-\tau} \phi_n(\zeta) d\zeta\right) \cdot \psi_n(t - \tau) d\tau. \tag{2.3.12}$$

The number of cells $N_n(t)$ in each generation as a function of time is given by

$$\begin{cases} N_0(t) = N(0) - \int_0^t \left(r_0^{div}(\tau) + r_0^{die}(\tau)\right) d\tau \\ N_n(t) = \int_0^t \left(2r_{n-1}^{div}(\tau) - r_n^{div} - r_n^{die}(\tau)\right) d\tau, \qquad n = 1, 2, 3, \cdots \end{cases} \tag{2.3.13}$$

In the previous work, [24], while doing the simulation we realized that fitting the cyton model to data from CFSE labeling is tremendous and time consuming especially in term of running time. Therefore we came out with a closed form easy to implement in the case where the waiting time to die and to divide is exponentially distributed. We obtained following model.

$$\begin{cases} r_0^{div}(t) = \dfrac{f_0 N(0)}{p_0} \cdot e^{-\gamma_0 t} \\ r_n^{div}(t) = \dfrac{2^n N(0)}{p_0 p^n} \cdot \left(\displaystyle\prod_{i=0}^{n} f_i\right) \cdot e^{-\gamma t} \left[\dfrac{e^{-(\gamma_0 - \gamma)t}}{(\gamma - \gamma_0)^n} - \displaystyle\sum_{i=0}^{n-1} \dfrac{t^i}{i!(\gamma - \gamma_0)^{n-i}} \right] \\ r_0^{die} = \dfrac{N(0)}{d} \left[(1 - f_0) e^{-t_0/d} + f_0 e^{-\gamma_0 t} \right] \\ r_n^{die} = \dfrac{2^n N(0)}{p_0 p^{n-1} d} \cdot \left(\displaystyle\prod_{i=0}^{n-1} f_i\right) \cdot \Bigg\{ (1 - f_n) \left[\dfrac{e^{-\gamma_0 t}}{(\frac{1}{d} - \gamma_0)(\gamma - \gamma_0)^{n-1}} - \displaystyle\sum_{i=0}^{n-2} \dfrac{e^{-\frac{t}{d}} \Gamma\left[i+1, (\gamma - \frac{1}{d})t\right]}{i!(\gamma - \gamma_0)^{n-1-i}(\gamma - \frac{1}{d})^{i+1}} \right] \\ \qquad + f_n \left[\dfrac{e^{-\gamma_0 t} - e^{-\gamma t}}{(\gamma - \gamma_0)^n} - \displaystyle\sum_{i=0}^{n-2} \dfrac{e^{-\gamma t} t^{i+1}}{(i+1)!(\gamma - \gamma_0)^{n-1-i}} \right] \Bigg\}, \end{cases} \tag{2.3.14}$$

where $\Gamma(s, x)$ is the lower incomplete gamma function defined by

$$\Gamma(s, x) = \int_0^x \xi^{s-1} e^{-\xi} d\xi, \tag{2.3.15}$$

p_0, p, d_0 and d are respectively the proliferation parameters, and the death parameters for exponential distributions corresponding to the 0th stage and other stages. For ample details on the construction of the model (2.3.14), refer to [24]

Even though the cyton model better capture the T cell proliferation dynamics than Smith-Martin model, Hyrien et al. [27] pointed the fact the model lies on the choice of the distributions to describe the two underlying process.

2.3.3 Age-dependent Multitype Braching Process

The Age-dependent Multitype Branching Process is a variant of branching processes developed by Harris. This type of model is famous for its good description of complex phenomena such as cell proliferation. The version we present here is that ofMiao et al. [28] . which is the result of critics of the ealier versions proposed by Hyrien and Zand [29], [30], [31]... The model is built on the event that occurs after a period corresponding to the duration of the cell cycle. As a result, Miao et al. [28] divides the cells according to these events:

- Type 1: Cells that will die
- Type 2: Cells that will stay quiescent
- Type 3: Cells that will divide.

Furthermore, the author introduced the following assumptions:

1. The experiments starts at time $t = 0$ with respectively $N^{(1)}(0)$ cells of type 1, $N^{(2)}(0)$ cells of type 2, and $N^{(3)}(0)$ cells of type 3;
2. Cells of any type from any generation evolve independently from others;
3. At the end of a cell lifetime, dying cells are eliminated from the system.
4. The lifetime of quiescent cells is beyond the length of the experiment.
5. Cells that are about to divide give two daughter cells of any of three types at the end of its lifetime. It is important to note that the two daughter cells can be of different types.
6. The lifetime of cells of the same type from the same generation follow the same distribution with the same distribution parameters.
7. The lifetime of cells of the same type but different generations may follow the same distribution with different distribution parameters.

As illustrated on the figure (2.3), the process start with an single progenitor of age zero which can be of any type. Since we do not know the type in which it belongs to, let P_1 be the probability of being of type 1, P_2 the probability of being of type 2 and $P_3 = 1 - P_1 - P_2$ the probability

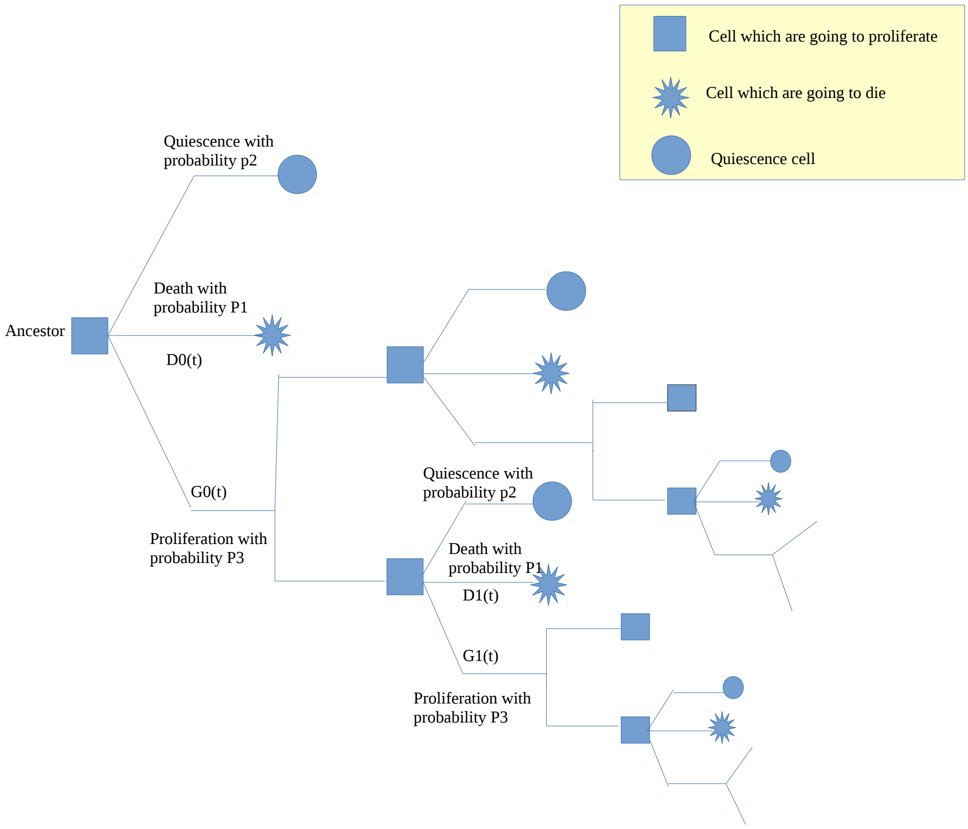

Figure 2.3: Description of branching process inspired from [28]

of being of type 3. The proliferation follow the type 3 branch. The lifetime distribution X_i of cells that are about to proliferate is represented by the cumulative distribution function (c.d.f) $G_i(t) = Pr(X_i \leq t)$ and the lifetime distribution T_i of cells that are going to die is represented by the c.d.f $D_i(t) = Pr(T_i \leq t)$ where the index i denotes the generation that the cell belongs to. By following the discussion by Miao et al. [28], which have been detailed by [24], we end up

$$\begin{cases} N_0^{(1)}(t) = N^{(1)}(0) \cdot (1 - D_0(t)) \\ N_0^{(2)}(t) = N^{(2)}(0) \\ N_0^{(3)}(t) = N^{(3)}(0) \cdot (1 - G_0(t)) \\ N_0^{(-1)}(t) = N^{(1)}(0) \cdot D_0(t) \\ N_n^{(1)}(t) = N^{(3)}(0) \cdot (2P_1)^n \cdot [G_0(t) * G^{*(n-1)}(t) - G_0(t) * G_1(t) * G^{*n-1}(t) * D(t)] \\ N_n^{(2)}(t) = N^{(3)}(0) \cdot (2P_2)^n \cdot [G_0(t) * G^{*(n-1)}(t)] \\ N_n^{(3)}(t) = N^{(3)}(0) \cdot (2P_3)^n \cdot [G_0(t) * G^{*(n-1)}(t) - G_0(t) * G_1(t) * G^{*n}(t)] \\ N_n^{(-1)}(t) = N^{(3)}(0) \cdot (2P_1)^n \cdot [G_0(t) * G^{*(n-1)}(t)]. \end{cases} \tag{2.3.16}$$

Although the presented model is robust and better estimates T cell dynamics, it remains difficult to implement and to analyze, meantime it is more favorable to the choice of one distribution than others ([27], [28]). Therefore a more flexible model is needed. Based on previous works, where the authors used the *Stochastic Master Equation* (SME) to describe similar processes. These works include the work of [32] where they used SME to describe cell division, Ndifon et al. [33] used the same approach to describe polymerase chain reaction and more recently Percus and Childress [34] used this approach to describe stem cell proliferation, it clearly appears that the SME approach would be the best way to describe T cell proliferation. In addition, Juwara [23] in his master thesis built a model that we are revisiting here. Furthermore this early model serves as basis for this work.

3. Mathematical Tools

This chapter is about presenting some mathematical tools which will be used in this work.

3.1 Markov Process

The following description of Markov process that we give here is not a full description but rather an introduction to material that will be used in the frame of this work. A better description can be found in [35].

Definition 2. A **stochastic process** is a sequence of events in which the outcome at any stage depends on some probability.

Definition 3. A **Markov process** is a continuous time stochastic process with the following properties:

1. The number of possible outcomes or states is finite.
2. The outcome at any stage depends only on the outcome of the previous stage.
3. The probabilities are constant over time.

   Therefore if $t_0 < t_1 < \cdots < t_n$ represents the instants on time scale then the set of random variables $\{X(t)\}$ whose state space $S = x_0, x_1, \cdots, x_p$ is said to follow a Markov process provided it holds the Markovian property:

$$P(x_k, t_k | x_{k-1}, t_{k-1}; x_{k-2}, t_{k-2}; \cdots ; x_0, t_0) = P(x_k, t_k | x_{k-1}, t_{k-1}) \tag{3.1.1}$$

   The property (3.1.1) is called Markov assumptions.

Before we move on, let us recall some results from probabilities theory:

Theorem 3.1. *Let $B_1, B_2, \cdots, B_k$ be a sequence of pairwise independent events, then we have*

(a) The joint probability is given by

$$P(A \cap B_i) = P(A|B_i)P(B_i) \tag{3.1.2}$$

(b) From (a) we can deduct

$$P(A) = \sum_{i=1}^{k} P(A|B_i)P(B_i) \tag{3.1.3}$$

From (3.1.1), (3.1.2), (3.1.3) we can establish **Chapman Kolmogrov Equation**:

$$P(A|C) = \sum_{i=1}^{k} P(A|B_i, C)P(B_i|C). \tag{3.1.4}$$

Among the Markov-like processes, there is a particular class known as Jump process. Jump process are used to model a discontinuous random variable over time. Since time is continuous, but the variable is discrete, the variable can represent a jump in a quantity or the occurrence of an event. In this work, the Jump process will be used to describe T-cell dynamics.

3.1.1 Jump process

Let $n_1, n_2, \cdots, n_k$ denote the possible state, $P_i(t)$ the probability to find the system of interest in state n_i at time t and $\omega(n_l \to n_p)dt$ the probability that the system of interest jump from the state n_l to the state n_p between t and $t + dt$. Then the probability to find the system in the state n_i at $t + dt$ is given by:

$$\begin{aligned} P_i(t+dt) &= P_i(t)Prob(\text{to stay in } n_i) + \sum_{j \neq i} P_j(t)Prob(\text{to jump in } n_i) \\ &= P_i(t)(1 - Prob(\text{to jump out of } n_i)) + \sum_{j \neq i} P_j(t)Prob(\text{to jump in } n_i) \\ &= P_i(t)(1 - \sum_{j \neq i} \omega(n_i \to n_j)dt + \sum_{j \neq i} P_j(t)\omega(n_i \to n_j)dt \end{aligned}$$

By using the definition of the derivative, we obtains

$$\frac{\partial P_i(t)}{\partial t} = \lim_{dt \to 0} \frac{P_i(t+dt) - P_i(t)}{dt} = \sum_{j \neq i} P_j(t)\omega(n_j \to n_i)dt - \sum_{j \neq i} P_i(t)\omega(n_i \to n_j)dt. \tag{3.1.5}$$

In (3.1.5), the first sum represents the probability to jump to n and the second sum the probability to jump out of n.

Numerous stochastic processes are characterized by probabilities laws that are difficult or even impossible to determine. However, the Probability generating function (PGF) can substitute the probability distribution, given that it carries the same information. Additionally, it allows extracting certain useful insights. In our case, PGF helps us to transform SME into the transport equation which is more digest. It also allows us to extract statistical quantities.

3.2 Probability Genereting Function and its properties

Definition 4. Let $X = (X_1, X_2, \cdots, X_n)$ be a multivariate discrete non negative random variable, The PGF associated with of X is defined as follow

$$G_X(S) = \mathbf{E}\left(S^X\right) = \sum_{X\in\mathbb{N}^n} S^X P(X) = \sum_{x_1,x_2,\cdots,x_n\geq 0}\prod_{i=1}^{n} S_i^{x_i} P(x_1, x_2, \cdots, x_n), \tag{3.2.1}$$

for all $S = (S_1, S_2, \cdots, S_n) \in \mathbb{R}^n$ for which the series (3.2.1) converges.

Remark 1. If $X_1, X_2, \cdots, X_p$ pairwise independent, then the multivariate PGF becomes

$$G_X(S) = \prod_{i=1}^{p} G_{X_i}(S_i) \tag{3.2.2}$$

As we mentioned earlier, the PGF is powerful and the advantages that will we use here is the fact that it allows us to derive the mean and the variance of the distribution without knowing explicitly the distribution. Hence we will compute the means, the variances and the covariances of cell counting. These results relies on the following results

- $G(\mathbf{1}, t) = \sum_{x_i=0}^{\infty} P(x_i, t) = 1$, here $\mathbf{1} = (1, 1, \cdots, 1)$
- the marginal mean is defined as the first moment at $S = (1, 1, \cdots, 1)$ and can be compute as follow
$$\left.\frac{\partial G(\boldsymbol{S}, t)}{\partial S_i}\right|_{\boldsymbol{S}=(1,1,\cdots,1)} = \mathbf{E}\left(X_i\right) \tag{3.2.3}$$
- In the same way we can be compute the second moment at $\boldsymbol{S} = 1$ as follow

$$\left.\frac{\partial^2 G(\boldsymbol{S}, t)}{\partial S_i^2}\right|_{\boldsymbol{S}=(1,1,\cdots,1)} = \mathbf{E}\left[X_i(X_i - 1)\right] \tag{3.2.4}$$

$$\left.\frac{\partial^2 G(\boldsymbol{S},t)}{\partial S_i \partial S_j}\right|_{\boldsymbol{S}=(1,1,\cdots,1)} = \mathbf{E}\left[X_i X_j\right] \tag{3.2.5}$$

- from (3.2.3) and (3.2.4) we can derive the variance since we know that

$$\mathbf{Var}\,(X_i) = \mathbf{E}\left[X_i(X_i-1)\right] + \mathbf{E}\,(X_i) - \mathbf{E}^{\,2}(X_i) \tag{3.2.6}$$

- from (3.2.3) and (3.2.5) we can derive the variance since we know that

$$\mathbf{Cov}\,(X_i, X_j) = \mathbf{E}\left[X_i X_j\right] - \mathbf{E}\,(X_i)\mathbf{E}\,(X_j) \tag{3.2.7}$$

3.3 Method of Characteristics

The method of characteristics allows us to transform the resolution of a first order PDE to the resolution of a system of ODEs by giving an explicit solution of the considered first other PDE. Let us see how it works by applying it to the transport equation. Let us consider the equation

$$\frac{\partial u(t,x)}{\partial t} + a(x)\cdot\nabla_x u(t,x) = 0, \tag{3.3.1}$$

where

$$a(x)\cdot\nabla_x u(x,t) = \sum_{i=1}^{n} a_i(x)\frac{\partial u(x,t)}{\partial x_i}, \quad a(x)\in\mathbb{R}^n\backslash\{0\}$$

and

$$\begin{aligned} u\colon \mathbb{R}_+ \times \mathbb{R}^n &\to \mathbb{R} \\ (t,x) &\mapsto u(t,x) \end{aligned}$$

In order to solve equation (3.3.1), one tries to find parametric curves

$$t = t(s), \quad x_i = x_i(s), \qquad i = \{1,2,\cdots,n\} \tag{3.3.2}$$

along which $u(x)$ remains constant. Such curves are given by the system of ODEs

$$\begin{cases} \dfrac{dx_i}{ds} = a_i(x), \qquad i = 1, 2, \cdots, n \\ \\ \dfrac{dt}{ds} = 1 \\ \\ \dfrac{du}{ds} = 0 \end{cases} \tag{3.3.3}$$

and they are called characteristic curves. Imposing the initial conditions

$$t(0) = t_0, \quad x_i(0) = x_{i_0} \qquad i = 1, 2, \cdots, n, \quad u(0) = u_0. \tag{3.3.4}$$

The solve of the combination of (3.3.3) and (3.3.4) allows one to determine the value of $u(x, t)$ at any point that lies on a characteristic curve through $x_0 = (x_{1_0}, x_{2_0}, \cdots, x_{n_0})$.

3.3.1 Maximum of Likelihood Estimation

Maximum Likelihood Estimation (MLE)is a technique widely used to estimate the parameters of a model. It allows to select the set of values of the model parameters that maximizes the likelihood function. That is to say, the approach maximizes the "agreement" of the selected model with the observed data. All is about to answer the following question: What are the parameters that maximize the probability of observing the particular sample $\boldsymbol{x} = (x_1, x_2, \cdots, x_n)$, assuming that a particular distribution with as yet unknown set of parameters $\boldsymbol{\theta} = (\theta_1, \theta_2, \cdots, \theta_m)$ generated the data? All we will do next will be to explain the method to answer this question, then we will apply this method to the studied case.

Let $X = (X_1, X_2, \cdots, X_n)$ be a sample of random variable chosen according to one of a family of probabilities. $Pr(\boldsymbol{x}|\theta)$ will be used to denote the density function for the data when θ is the true state of nature. The principle of maximum likelihood yields in a choice of the estimator $\hat{\theta}$ as the value for the parameter θ that makes the observed data most probable.

Definition 5. 1. The likelihood of a set of parameter values, $\boldsymbol{\theta} = (\theta_1, \theta_2, \cdots, \theta_m)$, given outcomes $\boldsymbol{x} = (x_1, x_2, \cdots, x_n)$, is equal to the probability of those observed outcomes given those parameter values, that is

$$L(\boldsymbol{\theta}|\boldsymbol{x}) = Pr(\boldsymbol{x}|\boldsymbol{\theta}) = Pr\big((X_1 = x_1) \cap (X_2 = x_2) \cap \cdots \cap (X_n = x_n)|\boldsymbol{\theta}\big). \tag{3.3.5}$$

2. The maximum likelihood estimator (MLE) is

$$\hat{\theta}(\boldsymbol{x}) = \arg\max_{\boldsymbol{\theta}} L(\boldsymbol{\theta}|\boldsymbol{x}) \tag{3.3.6}$$

Remark 2. 1. It is important to notice that in the expression (3.3.5) in the above definition, we are not dealing with a conditional probability given that the parameters $\theta_1, \theta_2, \cdots, \theta_m$ are not the outcomes of random variables.

2. In the case that the random variable $X_1, X_2, \cdots, X_n$ are independent and identically distributed (as in our case), the likelihood will be

$$L(\boldsymbol{\theta}|\boldsymbol{x}) = \prod_{i=1}^{n} Pr_{\boldsymbol{\theta}}\big(X_i = x_i\big). \tag{3.3.7}$$

Since we are dealing with a product, instead of maximizing the likelihood function, we maximize its logarithm (which is equivalent to (3.3.6)), that is to say

$$\hat{\theta}(\boldsymbol{x}) = \arg\max_{\boldsymbol{\theta}} \ln\left[L(\boldsymbol{\theta}|\boldsymbol{x})\right] = \arg\max_{\boldsymbol{\theta}} \sum_{i=1}^{n} \ln\left[Pr_{\boldsymbol{\theta}}(X_i = x_i)\right] \tag{3.3.8}$$

In many cases, the equation (3.3.8) has analytical solution, that is to say to have $\hat{\theta}$ explicitly as function of observed data. Let us give an example where the estimated parameter can be found analytically.

Example 1. A random variable of n values is collected from a negative binomial distribution with a given parameter k. Let us find the maximum Likelihood estimator $\hat{\pi}$ of the parameter π (the probability of success)

We have

$$P(x_i) = \binom{x_i - 1}{k - 1} \pi^k (1 - k)^{x_i - k},$$

the likelihood function is given by

$$L(\pi, x) = \prod_{i=1}^{n} \binom{x_i - 1}{k - 1} \pi^k (1 - k)^{x_i - k}$$

and the log-likelihood is

$$l(\pi, x) = \ln\big[L(\pi, x)\big] = \sum_{i=1}^{n} \Big[\ln\binom{x_i - 1}{k - 1} + k\ln\pi + (x_i - k)\ln(1-\pi)\Big].$$

$\hat{\pi}$ is the solution of the equation

$$\frac{\partial l(\pi, x)}{\partial \pi} = 0. \tag{3.3.9}$$

Let us find the solution of the above equation

$$\begin{aligned}
\frac{\partial l(\pi, x)}{\partial \pi} = 0 \Leftrightarrow & \sum_{i=1}^{n} \Big[\ln\binom{x_i - 1}{k - 1} + k\ln\pi + (x_i - k)\ln(1-\pi)\Big] = 0 \\
\Leftrightarrow & \frac{nk}{pi} - \frac{1}{1-\pi}\left(\sum_{i=1}^{n} x_i - nk\right) = 0 \\
\Leftrightarrow & nk - nk\pi = \pi\sum_{i=1}^{n} x_i - nk\pi \\
\Leftrightarrow & \pi = \frac{nk}{\sum_{i=1}^{n} x_i}.
\end{aligned} \tag{3.3.10}$$

Hence

$$\hat{\pi} = \frac{nk}{\sum_{i=1}^{n} x_i}.$$

In many cases as in the case studied in this work, it is difficult or even impossible to estimate analytically the desired parameters. In this case by using a computer algorithm, one can find or approach the needed parameters. Later we will present an algorithm to perform such a task.

4. Stochastic Master Equation

In this part, inspired by the work by Juwara [23], we construct a new model of T cell dynamic based on Markov Jump process. Additionally we derive some statistics, notably the means,the variances and covariances of cell counts.

4.1 Construction of the model

Let us consider a situation where newly produced T cells are in a quiescent phase of the cell cycle (G_0/G_1). These T cells are occasionally activated by external signals, notably antigens, including pathogen-derived or cytokines. T cells thus activated can enter the proliferation phase ($S/G_2/M$) of the cell cycle and then divide, producing two new daughter cells in a quiescent state and thereby exiting the cell cycle. Alternatively, the activated T cells can undergo activation-induced cell death. Now let us imagine that we have an *in vitro* T cell culture with conditions that mirror those encountered *in vivo* in the absence of specific T cell stimulation. At time t_0 we have N quiescent T cells. We provide a constant supply of low doses of antigen and cytokines that cause a subset of these cells to occasionally become activated and then enter phases $S/G_2/M$ of the cell cycle. As mentioned above, these activated cells can either divide, producing 2 daughter cells which are in the quiescent phase of the cell cycle, or die. We want to construct an SME describing the dynamics of this cell cycling process.

Let Q and A denote quiescent and activated cells respectively. Let the state of the T cell culture be described by the following vector $\boldsymbol{n} = (n_0^{(Q)}, n_1^{(Q)}, \cdots, n_h^{(Q)}, n_0^{(A)}, n_1^{(A)}, \cdots, n_h^{(A)})$, where $n_i^{(Q)}$ and $n_i^{(A)}$ are the numbers of type-Q cells and type-A cells respectively that have already divided i times, $0 \leq i \leq h$, and h is the so-called Hayflick limit [36]. The construction of the SMEs for the system lie on the probability density $P(\boldsymbol{n}, t)$ which represent the probability to find the system state $\boldsymbol{n}$ at time t. Let us recall the assumptions on which the model is built on:

- The system follows a discrete stochastic process and the state vector evolves randomly
- The system follows an instantaneous jump process between the possible states.

We define the step density function $P(\boldsymbol{n}, t|\boldsymbol{n}_0, t_0)$ which represent the probability to find the system at the state $\boldsymbol{n}$ at t given that it was at $\boldsymbol{n}_0$ at t_0. Let us remember that a cell divides,

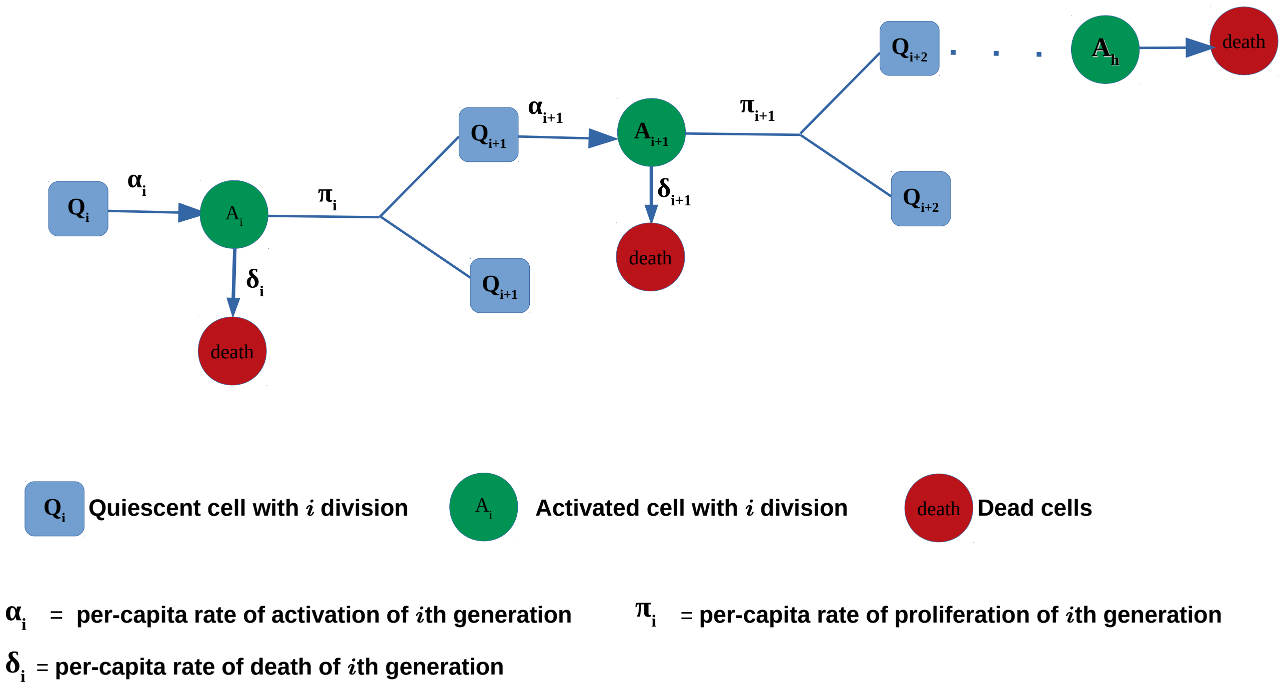

Figure 4.1: Description of the new model of T cell proliferation

remains quiescent or dies according a certain rule. We denote by $T(\boldsymbol{n}|\boldsymbol{n}')dt$ the probability that the system jump from the current state $\boldsymbol{n}'$ to $\boldsymbol{n}$ between t and $t+dt$ (dt is too small such that only one event can occur between t and $t+dt$). Therefore we can write the probability to find the system at the system at $\boldsymbol{n}$ at the time $t+dt$ is given by

$$\begin{aligned} P(\boldsymbol{n}, t+dt|\boldsymbol{n}', t) &= T(\boldsymbol{n}|\boldsymbol{n}')dt + 1 - Prob(\text{jump out } \boldsymbol{n}) \\ &= T(\boldsymbol{n}|\boldsymbol{n}')dt + \Big(1 - \sum_{\boldsymbol{m}} T(\boldsymbol{m}|\boldsymbol{n})dt\Big)\delta_{\boldsymbol{n},\boldsymbol{n}'} \end{aligned} \tag{4.1.1}$$

Where $(1 - \sum_{\boldsymbol{n}'} T(\boldsymbol{n}'|\boldsymbol{n}))\,\delta_{\boldsymbol{n},\boldsymbol{n}'}$ represents the probability of the system remain in his current state.

Let us illustrate how the formula holds by considering a system with only three possible states namely $\boldsymbol{n}_1$, $\boldsymbol{n}_2$ and $\boldsymbol{n}_3$. Let $\boldsymbol{n}_1$ be the state of interest, then (4.1.1) will be written as follow

$$\begin{aligned}
P(\boldsymbol{n}_1, t+dt|\boldsymbol{n}_1, t) &= T(\boldsymbol{n}_1|\boldsymbol{n}_1)dt + 1 - T(\boldsymbol{n}_1|\boldsymbol{n}_1)dt - T(\boldsymbol{n}_2|\boldsymbol{n}_1)dt - T(\boldsymbol{n}_3|\boldsymbol{n}_1)dt \\
&= 1 - (T(\boldsymbol{n}_2|\boldsymbol{n}_1)dt + T(\boldsymbol{n}_3|\boldsymbol{n}_1)dt) \\
&= 1 - Prob(\text{ to jump out of } \boldsymbol{n}_1 \text{ }) \\
&= Prob(\text{ to stay at } \boldsymbol{n}_1)
\end{aligned}$$

By using the same reasoning, we have

$$P(\boldsymbol{n}_1, t+dt|\boldsymbol{n}_2, t) = T(\boldsymbol{n}_1|\boldsymbol{n}_2)dt, \tag{4.1.2}$$

and

$$P(\boldsymbol{n}_1, t+dt|\boldsymbol{n}_3, t) = T(\boldsymbol{n}_1|n_3)dt, \tag{4.1.3}$$

which can be generalize for a system of k states ($k \in \mathbb{N}$).

By using the above theorem and the Chapman Kolmogrov equation, we can now built the SME for the system.

$$\begin{aligned}
P(\boldsymbol{n}, t+dt|\boldsymbol{n}_0, t_0) &= \sum_{\boldsymbol{m}} P(\boldsymbol{n}, t+dt|\boldsymbol{m}, t, \boldsymbol{n}_0, t_0)P(\boldsymbol{m}, t|\boldsymbol{n}_0, t_0) \\
&= \sum_{\boldsymbol{m}} P(\boldsymbol{n}, t+dt|\boldsymbol{m}, t)P(\boldsymbol{m}, t|\boldsymbol{n}_0, t_0) \\
&= \sum_{\boldsymbol{m}} \left[T(\boldsymbol{n}|\boldsymbol{m})dt + \left(1 - \sum_{\boldsymbol{n}'} T(\boldsymbol{n}'|\boldsymbol{n})dt\right)\delta_{\boldsymbol{n},\boldsymbol{m}} \right] P(\boldsymbol{m}, t|\boldsymbol{n}_0, t_0) \\
&= \sum_{\boldsymbol{m}} T(\boldsymbol{n}|\boldsymbol{m})dtP(\boldsymbol{m}, t|\boldsymbol{n}_0, t_0) + \sum_{\boldsymbol{m}} \left(1 - \sum_{\boldsymbol{n}'} T(\boldsymbol{n}'|\boldsymbol{n})dt\right)\delta_{\boldsymbol{n},\boldsymbol{m}} P(\boldsymbol{m}, t|\boldsymbol{n}_0, t_0) \\
&= \sum_{\boldsymbol{m}} T(\boldsymbol{n}|\boldsymbol{m})dtP(\boldsymbol{m}, t|\boldsymbol{n}_0, t_0) + \left(1 - \sum_{\boldsymbol{n}'} T(\boldsymbol{n}'|\boldsymbol{n})dt\right) P(\boldsymbol{n}, t|\boldsymbol{n}_0, t_0).
\end{aligned} \tag{4.1.4}$$

as $\left(1 - \sum_{\boldsymbol{n}'} T(\boldsymbol{n}'|\boldsymbol{n})dt\right)\delta_{\boldsymbol{n},\boldsymbol{m}} = 0$ when $\boldsymbol{m} \neq \boldsymbol{n}$.

Therefore, we have

$$\frac{P(\boldsymbol{n}, t+dt|\boldsymbol{n}_0, t_0) - P(\boldsymbol{n}, t|\boldsymbol{n}_0, t_0)}{dt} = \sum_{\boldsymbol{m}} T(\boldsymbol{n}|\boldsymbol{m})P(\boldsymbol{m}, t|\boldsymbol{n}_0, t_0) - \sum_{\boldsymbol{m}} T(\boldsymbol{m}|\boldsymbol{n})P(\boldsymbol{n}, t|\boldsymbol{n}_0, t_0)$$

$$= \sum_{\boldsymbol{m}\neq\boldsymbol{n}} \left[T(\boldsymbol{n}|\boldsymbol{m})P(\boldsymbol{m},t|\boldsymbol{n}_0,t_0) - T(\boldsymbol{m}|\boldsymbol{n})P(\boldsymbol{n},t|\boldsymbol{n}_0,t_0)\right], \tag{4.1.5}$$

by tending dt towards 0, we obtain

$$\frac{\partial P(\boldsymbol{n},t|\boldsymbol{n}_0,t_0)}{\partial t} = \sum_{\boldsymbol{m}\neq\boldsymbol{n}} \left[T(\boldsymbol{n}|\boldsymbol{m})P(\boldsymbol{m},t|\boldsymbol{n}_0,t_0) - T(\boldsymbol{m}|\boldsymbol{n})P(\boldsymbol{n},t|\boldsymbol{n}_0,t_0)\right]. \tag{4.1.6}$$

Given that we know that the system was at the state $\boldsymbol{n}_0$ at t_0, it is not really necessary to state it. Therefore, (4.1.6) becomes

$$\frac{\partial P(n,t)}{\partial t} = \sum_{\boldsymbol{m}\neq\boldsymbol{n}} \left[T(\boldsymbol{n}|\boldsymbol{m})P(\boldsymbol{m},t) - T(\boldsymbol{m}|\boldsymbol{n})P(\boldsymbol{n},t)\right], \tag{4.1.7}$$

which tell us that the rate of change of the system from a given state $\boldsymbol{n}$ over the time, is equal to sum of all the possible jump to $\boldsymbol{n}$ minus the sum of all of possible jump out of $\boldsymbol{n}$.

As we have a fully description of the system in term of SME, let us go further by describing in details the possible events that lead to a jump into $\boldsymbol{n}$:

1. A quiescent cell with i divisions becomes activated and enters state A: This event causes a transition into $\boldsymbol{n}$ from the initial state $\boldsymbol{n}' = (\cdots, n_i^{(Q)}+1, \cdots, n_i^{(A)}-1, \cdots)$ with the per-capita rate α_i.
2. An activated cell with i divisions, $i < h$, divides, this causes a transition into state $\boldsymbol{n}$ from the initial state $\boldsymbol{n}' = (\cdots, n_{i+1}^{(Q)}-2, \cdots, n_i^{(A)}+1, \cdots)$, with the per-capita rate π_i.
3. An activated cell with i divisions dies and causes a transition into state $\boldsymbol{n}$ from the initial state $\boldsymbol{n}' = (\cdots, n_i^{(Q)}, \cdots, n_i^{(A)}+1, \cdots)$, with the per-capita rate δ_i.

The possible events that cause a jump out of the $\boldsymbol{n}$ are the same, namely activation, division and death.

Note that the equation (4.1.7) describes the changes within the state $\boldsymbol{n}$ over the time. I can be split in to two descriptions namely a positive part corresponding jumping into $\boldsymbol{n}$ and negative part corresponding to jumping out of $\boldsymbol{n}$. By following these argument we end with

$$\begin{aligned}\frac{\partial P(\boldsymbol{n},t)}{\partial t} \;=\;& \sum_{i=0}^{h}\alpha_i(n_i^{(Q)}+1)P(\cdots,n_i^{(Q)}+1,\cdots,n_i^{(A)}-1,\cdots,t)\\ &+\sum_{i=0}^{h-1}\pi_i(n_i^{(A)}+1)P(\cdots,n_{i+1}^{(Q)}-2,\cdots,n_i^{(A)}+1,\cdots,t)\\ &+\sum_{i=0}^{h}\delta_i(n_i^{(A)}+1)P(\cdots,n_i^{(Q)},\cdots,n_i^{(A)}+1,\cdots,t)\\ &-\sum_{i=0}^{h}(\alpha_i n_i^{(Q)}+\delta_i n_i^{(A)})P(\boldsymbol{n},t)-\sum_{i=0}^{h-1}\pi_i n_i^{(A)}P(\boldsymbol{n},t)\end{aligned} \tag{4.1.8}$$

As we mentioned it before, one of the main goals of this essay is to compute the moments of cell counting. in order to do so, we will make use of the powerful combinatorial item called the *probability generating function* (PGF) as defined in the preliminaries.

The PGF is useful when dealing with discrete random variables. Its particular strength is that it gives us an easy way to derive the distribution associated with the sum of two independent random variables. As we announced in the Mathematical tools, here we are dealing with the multivariate distribution made of independent random variable. To be able to proceed, we will adapt what we saw in the previous section to the case studied here.

$$G(\boldsymbol{S},t)=\sum_{n\in\Lambda}\boldsymbol{S}^{\boldsymbol{n}}P(\boldsymbol{n},t), \tag{4.1.9}$$

where

$$\begin{cases}\boldsymbol{S}=(Q_0,Q_1,\cdots,Q_h,A_0,A_1,\cdots,A_h)\\ \boldsymbol{n}=(n_0^{(Q)},n_1^{(Q)},\cdots,n_h^{(Q)},n_0^{(A)},n_1^{(A)},\cdots,n_h^{(A)})\\ \Lambda=\mathbb{N}^{2h+2}.\end{cases}$$

and

$$\boldsymbol{S}^{\boldsymbol{n}}=\prod_{i=0}^{h}(Q_i)^{n_i^{(Q)}}\cdot\prod_{i=0}^{h}(A_i)^{n_i^{(A)}}. \qquad \text{(just a notation)}$$

To solve the SME, we transform it to equation where there new unknown is the PGF. So, we multiply (4.1.8) by $\boldsymbol{S}^{\boldsymbol{n}}$, and by summing over the possible values of $\boldsymbol{n}$ and by interchanging the summation and the derivative, one obtains

$$
\begin{aligned}
\frac{\partial \sum\limits_{\boldsymbol{n}\in\Lambda} \boldsymbol{S}^{\boldsymbol{n}} P(\boldsymbol{n},t)}{\partial t} = & \sum_{i=0}^{h} \alpha_i \sum_{\boldsymbol{n}\in\Lambda} \boldsymbol{S}^{\boldsymbol{n}} (n_i^{(Q)}+1) P(\cdots, n_i^{(Q)}+1, \cdots, n_i^{(A)}-1, \cdots, t) \\
& + \sum_{i=0}^{h-1} \pi_i \sum_{\boldsymbol{n}\in\Lambda} \boldsymbol{S}^{\boldsymbol{n}} (n_i^{(A)}+1) P(\cdots, n_{i+1}^{(Q)}-2, \cdots, n_i^{(A)}+1, \cdots, t) \\
& + \sum_{i=0}^{h} \delta_i \sum_{\boldsymbol{n}\in\Lambda} \boldsymbol{S}^{\boldsymbol{n}} (n_i^{(A)}+1) P(\cdots, n_i^{(Q)}, \cdots, n_i^{(A)}+1, \cdots, t) \\
& - \sum_{i=0}^{h} \sum_{\boldsymbol{n}\in\Lambda} \boldsymbol{S}^{\boldsymbol{n}} (\alpha_i n_i^{(Q)} + \delta_i n_i^{(A)}) P(\boldsymbol{n},t) - \sum_{i=0}^{h-1} \pi_i \sum_{\boldsymbol{n}\in\Lambda} \boldsymbol{S}^{\boldsymbol{n}} n_i^{(A)} P(\boldsymbol{n},t)
\end{aligned}
\tag{4.1.10}
$$

To simplify the SME, we will compute each term appearing in the right-side of the equation (4.1.10) by using power series rules of computation, that leads us to

$$
\begin{aligned}
& \sum_{i=0}^{h} \alpha_i \sum_{\boldsymbol{n}\in\Lambda} \boldsymbol{S}^{\boldsymbol{n}} (n_i^{(Q)}+1) P(\cdots, n_i^{(Q)}+1, \cdots, n_i^{(A)}-1, \cdots, t) \\
= & \sum_{i=0}^{h} \alpha_i \sum_{n_0^{(Q)}=0}^{\infty} \cdots \sum_{n_i^{(Q)}=0}^{\infty} \cdots \sum_{n_h^{(Q)}=0}^{\infty} \sum_{n_0^{(A)}=0}^{\infty} \cdots \sum_{n_i^{(A)}=1}^{\infty} \cdots \sum_{n_h^{(A)}=0}^{\infty} \left(\prod_{\substack{j=0 \\ j\neq i}}^{h} Q_j^{n_j^{(Q)}} A_j^{n_j^{(A)}} \right) Q_i^{n_i^{(Q)}} A_i^{n_i^{(A)}} \\
& (n_i^{(Q)}+1) P(\cdots, n_i^{(Q)}+1, \cdots, n_i^{(A)}-1, \cdots, t) \\
= & \sum_{i=0}^{h} \alpha_i A_i \sum_{n_0^{(Q)}=0}^{\infty} \cdots \sum_{n_i^{(Q)}=0}^{\infty} \cdots \sum_{n_h^{(Q)}=0}^{\infty} \sum_{n_0^{(A)}=0}^{\infty} \cdots \sum_{n_i^{(A)}=1}^{\infty} \cdots \sum_{n_h^{(A)}=0}^{\infty} \left(\prod_{\substack{j=0 \\ j\neq i}}^{h} Q_j^{n_j^{(Q)}} A_j^{n_j^{(A)}} \right) Q_i^{n_i^{(Q)}} A_i^{n_i^{(A)}-1} \\
& (n_i^{(Q)}+1) P(\cdots, n_i^{(Q)}+1, \cdots, n_i^{(A)}-1, \cdots, t) \\
= & \sum_{i=0}^{h} \alpha_i \sum_{n_0^{(Q)}=0}^{\infty} \cdots \sum_{n_i^{(Q)}=1}^{\infty} \cdots \sum_{n_h^{(Q)}=0}^{\infty} \sum_{n_0^{(A)}=0}^{\infty} \cdots \sum_{n_i^{(A)}=0}^{\infty} \cdots \sum_{n_h^{(A)}=0}^{\infty} \left(\prod_{\substack{j=0 \\ j\neq i}}^{h} Q_j^{n_j^{(Q)}} A_j^{n_j^{(A)}} \right) Q_i^{n_i^{(Q)}-1} A_i^{n_i^{(A)}} \\
& n_i^{(Q)} P(\cdots, n_i^{(Q)}, \cdots, n_i^{(A)}, \cdots, t) \\
= & \sum_{i=0}^{h} \alpha_i A_i \frac{\partial G(\boldsymbol{S},t)}{\partial Q_i}.
\end{aligned}
$$

Using the same approach, we have the following equalities:

$$\sum_{i=0}^{h-1} \pi_i \sum_{\boldsymbol{n}\in\Lambda} \boldsymbol{S}^{\boldsymbol{n}} (n_i^{(A)}+1) P(\cdots, n_{i+1}^{(Q)}-2, \cdots, n_i^{(A)}+1, \cdots, t) = \sum_{i=0}^{h-1} \pi_i Q_{i+1}^2 \frac{\partial G(\boldsymbol{S},t)}{\partial A_i} \tag{4.1.11}$$

$$\sum_{i=0}^{h} \delta_i \sum_{\boldsymbol{n}\in\Lambda} \boldsymbol{S}^{\boldsymbol{n}} (n_i^{(A)}+1) P(\cdots, n_{i}^{(Q)}, \cdots, n_i^{(A)}+1, \cdots, t) = \sum_{i=0}^{h} \delta_i \frac{\partial G(\boldsymbol{S},t)}{\partial A_i}, \tag{4.1.12}$$

$$\sum_{i=0}^{h} \sum_{\boldsymbol{n}\in\Lambda} \boldsymbol{S}^{\boldsymbol{n}} (\alpha_i n_i^{(Q)} + \delta_i n_i^{(A)}) P(\boldsymbol{n}, t) = \sum_{i=0}^{h} \alpha_i Q_i \frac{\partial G(\boldsymbol{S},t)}{\partial Q_i} + \sum_{i=0}^{h} \delta_i A_i \frac{\partial G(\boldsymbol{S},t)}{\partial A_i}, \tag{4.1.13}$$

and

$$\sum_{i=0}^{h-1} \pi_i \sum_{\boldsymbol{n}\in\Lambda} \boldsymbol{S}^{\boldsymbol{n}} n_i^{(A)} P(\boldsymbol{n}, t) = \sum_{i=0}^{h-1} \pi_i A_i \frac{\partial G(\boldsymbol{S},t)}{\partial A_i}. \tag{4.1.14}$$

By replacing the above equation into (4.1.10) and by rearranging, the SME equation becomes:

$$\begin{aligned} \frac{\partial G(\boldsymbol{S},t)}{\partial t} &= \sum_{i=0}^{h} \alpha_i (A_i - Q_i) \frac{\partial G(\boldsymbol{S},t)}{\partial Q_i} + \sum_{i=0}^{h-1} \left[\delta_i (1-A_i) + \pi_i \left(Q_{i+1}^2 - A_i \right) \right] \frac{\partial G(\boldsymbol{S},t)}{\partial A_i} \\ &\quad + \delta_h (1-A_h) \frac{\partial G(\boldsymbol{S},t)}{\partial A_h} \end{aligned} \tag{4.1.15}$$

Given that we started the process with $n_0^{(Q)}$ cells, then our initial state is therefore

$$\boldsymbol{n}_0 = (n_0^{(Q)}, n_1^{(Q)} = 0, \cdots, n_h^{(Q)} = 0, n_0^{(A)} = 0, n_1^{(A)} = 0, \cdots, n_h^{(A)} = 0),$$

Furthermore, we are sure to have the system is at the state $\boldsymbol{n}_0$ at $t=0$, then we can express the probability at $t=0$ as follow

$$P(\boldsymbol{n}, 0) = \begin{cases} 1 & \text{if } \boldsymbol{n} = \boldsymbol{n}_0 \\ 0 & \text{if } \boldsymbol{n} \neq \boldsymbol{n}_0 \end{cases} = \delta_{\boldsymbol{n}_0, \boldsymbol{n}},$$

which allows us to have the initial condition of the SME

$$G(\boldsymbol{S},0) = \sum_{\boldsymbol{n}\in\Lambda} \boldsymbol{S}^{\boldsymbol{n}} P(\boldsymbol{n},=0) = \sum_{\boldsymbol{n}\in\Lambda} \boldsymbol{S}^{\boldsymbol{n}} \cdot \delta_{\boldsymbol{n}_0,\boldsymbol{n}} = \boldsymbol{S}^{\boldsymbol{n}_0} = Q_0^{n_0^{(Q)}}. \tag{4.1.16}$$

By looking closely equation (4.1.15), it turn out it correspond to a first order linear differential equation know as transport equation, and therefore can be solved analytically using the methods of characteristics.

4.2 Solution of the Stochastic Master equation

To solve the obtained PDE, we use the method of characteristic that we earlier gave a brief description.

The system (4.1.15) can be written as

$$\begin{aligned}
&\frac{\partial G(\boldsymbol{S},t)}{\partial t} - \sum_{i=0}^{h} \alpha_i (A_i - Q_i) \frac{\partial G(\boldsymbol{S},t)}{\partial Q_i} - \sum_{i=0}^{h-1} \left[\delta_i(1-A_i) + \pi_i \left(Q_{i+1}^2 - A_i\right)\right] \frac{\partial G(\boldsymbol{S},t)}{\partial A_i} \\
&-\delta_h(1-A_h)\frac{\partial G(\boldsymbol{S},t)}{\partial A_h} = 0,
\end{aligned} \tag{4.2.1}$$

by applying the method of characteristics, one obtains the following system of equations

$$\frac{dQ_i}{dx} = \alpha_i(A_i - Q_i), \quad i = 0,1,\cdots,h \tag{4.2.2}$$

$$\frac{dA_i}{dx} = -\left[\delta_i(1-A_i) + \pi_i\left(Q_{i+1}^2 - A_i\right)\right], \quad i = 0,1,\cdots,h-1 \tag{4.2.3}$$

$$\frac{dA_h}{dx} = -\delta_h(1-A_h) \tag{4.2.4}$$

$$\frac{dt}{dx} = 1 \tag{4.2.5}$$

$$\frac{dG}{dx} = 0, \tag{4.2.6}$$

with the initial condition

$$G(\boldsymbol{S},0) = (Q_0)^{n_0^{(Q)}} \tag{4.2.7}$$

It is theoretically possible to have the expressions of Q_i and A_i as function of x and therefore the expression of $G(\boldsymbol{S}(t), t)$ through the following algorithm:

1. From (4.2.5), set $x = t$.
2. Solve (4.2.4) and obtain A_h.
3. Replace A_h in (4.2.2) and solve to obtain Q_h.
4. Replace Q_h in (4.2.3) to obtain A_{h-1}.
5. Repeat the steps 2., 3., 4., until you obtain the value of Q_0.

In practice, the above algorithm is hard to be applied especially when $h > 2$. Indeed as we can notice the system can be solved using a backward recursion and after two steps, the number of parameters in the system explode. To show how tremendous it is, let us compute A_h, Q_h, A_{h-1} and Q_{h-1}.

from (4.2.4), we have

$$\begin{aligned} \frac{dA_h}{dt} - \delta_h A_h = -\delta_h \quad &\Leftrightarrow \quad \frac{dA_h e^{-\delta_h t}}{dt} = -\delta_h e^{-\delta_h t} \\ &\Rightarrow \quad A_h(t) = 1 - e^{\delta_h t} + g_h e^{-\delta_h t}, \quad g_h = A_h(0). \end{aligned} \tag{4.2.8}$$

From (4.2.2), one obtains

$$\begin{aligned} \frac{dQ_h}{dt} - \alpha_h Q_h = -\alpha_h A_h \quad &\Leftrightarrow \quad \frac{dQ_h e^{-\alpha_h t}}{dt} = -\alpha_h \left(e^{-\alpha_h t} + (g_h - 1)e^{(\delta_h - \alpha_h)t}\right) \\ &\Rightarrow \quad Q_h(t) = 1 + (f_h - 1)e^{\alpha_h t} + \frac{\delta_h (g_h - 1)}{\alpha_h - \delta_h}\left(e^{\delta_h t} - e^{\alpha_h t}\right), \quad f_h = Q_h(0). \end{aligned} \tag{4.2.9}$$

From (4.2.3) it follows that

$$\begin{aligned} \frac{dA_{h-1}}{dt} - (\delta_{h-1} + \pi_{h-1})A_{h-1} &= -\delta_{h-1} - \pi_{h-1} Q_h^2 \\ &= -\delta_{h-1} - \pi_{h-1}\left(1 + (f_h - 1)e^{\alpha_h t} + \frac{\delta_h (g_h - 1)}{\alpha_h - \delta_h}(e^{\delta_h t} - e^{\alpha_h t})\right)^2 \end{aligned}$$

$$= \quad -\beta_{h-1} - \pi_{h-1}\left(2p_1 e^{\alpha_h t} + p_1^2 e^{2\alpha_h t} + 2p_2 e^{(\delta+\alpha_h)t} + p_3 e^{2\delta_h t}\right),$$

with

$$\begin{cases} p_1 = f_h - 1 - \dfrac{\delta_h(g_h-1)}{\alpha_h-\delta_h} \\ p_2 = \dfrac{\delta_h(g_h-1)}{\alpha_h-\delta_h} p_1 \\ p_3 = \left(\dfrac{\delta_h(g_h-1)}{\alpha_h-\delta_h}\right)^2 \\ \beta_{h-1} = \delta_{h-1} + \pi_{h-1} \end{cases}$$

Hence, the above differential equation is equivalent to

$$\begin{aligned} \frac{dA_{h-1}e^{-\beta_{h-1}}}{dt} &= -\beta_{h-1}e^{-\beta_{h-1}} - \pi_{h-1}e^{-\beta_{h-1}}\left(2p_1 e^{\alpha_h t} + p_1^2 e^{2\alpha_h t} + 2p_2 e^{(\delta+\alpha_h)t} + p_3 e^{2\delta_h t}\right) \\ &= -\beta_{h-1}e^{-\beta_{h-1}} - \pi_{h-1}\left(2p_1 e^{(\alpha_h-\beta_{h-1})t} + p_1^2 e^{(2\alpha_h-\beta_{h-1})t} + 2p_2 e^{(\delta_h+\alpha_h-\beta_{h-1})t}\right. \\ &\quad \left. + p_3 e^{(2\delta_h-\beta_{h-1})t}\right) \end{aligned}$$

Therefore,

$$\begin{aligned} A_{h-1}(t) &= (1-e^{\beta_{h-1}t}) - \frac{2\pi_{h-1}p_1}{\alpha_h-\beta_{h-1}}(e^{\alpha_h t}-e^{\beta_{h-1}t}) - \frac{\pi_{h-1}p^2}{2\alpha_h-\beta_{h-1}}(e^{2\alpha_h t}-e^{\beta_{h-1}t}) \\ &\quad -\frac{2\pi_{h-1}p_2}{\delta_h+\alpha_h-\beta_{h-1}}(e^{(\delta_h+\alpha_h)t}-e^{\beta_{h-1}t}) - \frac{\pi_{h-1}p_3}{2\delta_h-\beta_{h-1}}(e^{2\delta_h t}-e^{\beta_{h-1}t}) + g_{h-1}e^{\beta_{h-1}t} \\ &= 1 + (g_{h-1}-1+m_1+m_2+m_3+m_4)e^{\beta_{h-1}t} - m_1 e^{\alpha_h t} - m_2 e^{2\alpha_h t} - m_3 e^{(\delta_h+\alpha_h)t} \\ &\quad -m_4 e^{2\delta_h t}, \end{aligned} \tag{4.2.10}$$

where

$$\begin{cases} m_1 = \dfrac{2\pi_{h-1}p_1}{\alpha_h-\beta_{h-1}} \\ m_2 = \dfrac{\pi_{h-1}p^2}{2\alpha_h-\beta_{h-1}} \\ m_3 = \dfrac{2\pi_{h-1}p_2}{\delta_h+\alpha_h-\beta_{h-1}} \\ m_4 = \dfrac{\pi_{h-1}p_3}{2\delta_h-\beta_{h-1}}. \end{cases}$$

From (4.2.2), it follows that

$$\frac{dQ_{h-1}}{dt} - \alpha_{h-1}Q_{h-1} = -\alpha_{h-1}A_{h-1},$$

which is equivalent to

$$
\begin{aligned}
\frac{d(Q_{h-1}e^{-\alpha_{h-1}t})}{dt} &= -\alpha_{h-1}\left(e^{-\alpha_{h-1}t} + Te^{(\beta_{h-1}-\alpha_{h-1})t} - m_1 e^{(\alpha_h-\alpha_{h-1})t} - m_2 e^{(2\alpha_h-\alpha_{h-1})t}\right. \\
&\quad \left. -m_3 e^{(\delta_h+\alpha_h-\alpha_{h-1})t} - m4 e^{(2\delta_h-\alpha_{h-1})t}\right),
\end{aligned}
$$

which implies that

$$
\begin{aligned}
Q_{h-1} &= \alpha_{h-1}e^{\alpha_{h-1}t}\left[-\frac{1}{\alpha_{h-1}}(e^{-\alpha_{h-1}t}-1) + \frac{T}{\beta_{h-1}-\alpha_{h-1}}(e^{(\beta_{h-1}-\alpha_{h-1})t}-1)\right. \\
&\quad -\frac{m_1}{\alpha_h-\alpha_{h-1}}(e^{(\alpha_h-\alpha_{h-1})t}-1) - \frac{m_2}{2\alpha_h-\alpha_{h-1}}(e^{(2\alpha_h-\alpha_{h-1})t}-1) \\
&\quad \left. -\frac{m_3}{\delta_h+\alpha_h-\alpha_{h-1}}(e^{(\delta_h\alpha_h-\alpha_{h-1})t}-1) - \frac{m_4}{2\delta_h-\alpha_{h-1}}(e^{(2\delta_h-\alpha_{h-1})t}-1)\right] + f_{h-1}e^{\alpha_{h-1}t} \\
&= 1 + (f_{h-1}-1+l_1-l_2-l_3-l_4-l_5)\,e^{\alpha_{h-1}t} - l_1 e^{\beta_{h-1}t} + l_2 e^{\alpha_h t} + l_3 e^{2\alpha_h t} \\
&\quad + l_4 e^{(\delta_h+\alpha_h)t} + l_5 e^{\delta_h t}, \qquad (4.2.11)
\end{aligned}
$$

with

$$
\begin{cases}
T = g_{h-1} - 1 + m_1 + m_2 + m_3 + m_4 \\
l_1 = \dfrac{\alpha_{h-1}T}{\beta_{h-1}-\alpha_{h-1}} \\
l_2 = \dfrac{\alpha_{h-1}m_1}{\alpha_h-\alpha_{h-1}} \\
l_3 = \dfrac{\alpha_{h-1}m_2}{2\alpha_h-\alpha_{h-1}} \\
l_4 = \dfrac{\alpha_{h-1}m_3}{\delta_h\alpha_h-\alpha_{h-1}} \\
l_5 = \dfrac{\alpha_{h-1}m_4}{2\delta_h-\alpha_{h-1}}.
\end{cases}
$$

Even though is difficult to have a general formula for $G(\boldsymbol{S}, t)$, we can still extract some useful information such as the means the variances and the covariances of cell counts. This will be the subject of the next section.

4.3 Mean, Variance of cell counts in the case where activation probability is constant

4.3.1 Mean of cell counts

As we mentioned in chapter 3, from the probability generating function of a multivariate random variable, we can compute the marginal mean. In our case we want to derive the means $\mathbf{E}\left(n_i^{(Q)}\right)$ and $\mathbf{E}\left(n_i^{(A)}\right)$ for each generation i, $i = 0, 1, \cdots, h$.

By definition,

$$\mathbf{E}\left(n_i^{(Q)}\right) = \left.\frac{\partial G(\boldsymbol{S}, t)}{\partial Q_i}\right|_{\boldsymbol{S}=(1,1,\cdots,1)}, \tag{4.3.1}$$

and

$$\mathbf{E}\left(n_i^{(A)}\right) = \left.\frac{\partial G(\boldsymbol{S}, t)}{\partial A_i}\right|_{\boldsymbol{S}=(1,1,\cdots,1)} \tag{4.3.2}$$

Let us compute each of this mean. To ease the computation, let us introduce the following notations

$$f_i = Q_i(0) \quad \text{and} \quad g_i = A_i(0).$$

The mean of quiescent cells undergone i division can be computed as follow

$$\begin{aligned}
\mathbf{E}\left(n_i^{(Q)}\right) &= \left.\frac{\partial G(\boldsymbol{S}, t)}{\partial Q_i}\right|_{\boldsymbol{S}=(1,1,\cdots,1)} \\
&= \left.\frac{\partial Q_0^{n_0^{(Q)}}}{\partial Q_i}\right|_{\boldsymbol{S}=(1,1,\cdots,1)} \\
&= n_0^{(Q)} Q_0^{n_0^{(Q)}-1} \left.\frac{\partial Q_0}{\partial Q_i}\right|_{\boldsymbol{S}=(1,1,\cdots,1)} \\
&= n_0^{(Q)} Q_0^{n_0^{(Q)}-1} \frac{\partial Q_0}{\partial f_0} \left(\prod_{j=0}^{i-1} \frac{\partial f_j}{\partial g_j} \frac{\partial g_j}{\partial f_{j+1}}\right) \left.\frac{\partial f_i}{\partial Q_i}\right|_{\boldsymbol{S}=(1,1,\cdots,1)}.
\end{aligned} \tag{4.3.3}$$

Let us compute each of terms appearing in (4.3.3). we have

$$\frac{\partial Q_0}{\partial f_0} = e^{\alpha_0 x} \quad \text{and} \quad \frac{\partial f_i}{\partial Q_i} = e^{-\alpha_i x}.$$

Now let us find the expressions of $\frac{\partial f_j}{\partial g_j}$ and $\frac{\partial g_j}{\partial f_{j+1}}$.

From (4.2.2), one obtains

$$Q_i(x) = e^{\alpha_i x}\left(f_i - \alpha_i \int_0^x e^{-\alpha_i s} A_i(s)ds\right). \tag{4.3.4}$$

By setting $\beta_i = \delta_i + \pi_i$, it follow from (4.2.3) that

$$A_i(x) = e^{\beta_i x}\left[g_i + \frac{\delta_i}{\beta_i}(e^{\beta_i x} - 1) - \pi_i \int_0^x e^{\beta_i s} Q_{i+1}^2(s)ds\right]. \tag{4.3.5}$$

Replacing (4.3.4) in (4.3.5) with have

$$\begin{aligned} Q_i(x) &= e^{\alpha_i x}\left[f_i - \frac{\alpha_i g_i}{\beta_i - \alpha_i}(e^{(\beta_i-\alpha_i)x} - 1)\right] - \frac{\alpha_i \delta_i}{\beta_i} e^{\alpha_i x} \int_0^x (e^{-\alpha_i s} - e^{(\beta-\alpha_i)s})ds \\ &\quad + \alpha_i \pi_i e^{\alpha_i x} \int_0^x e^{(\beta_i-\alpha_i)s} \int_0^s e^{-\beta_i \tau} Q_{i+1}^2(\tau)d\tau ds \end{aligned}$$

Since $Q_i(x)$ does not depend on g_i,

$$\frac{\partial Q_i(x)}{\partial g_i} = e^{\alpha_i x}\left[\frac{\partial f_i}{\partial g_i} - \frac{\alpha_i}{\beta_i - \alpha_i}(e^{(\beta_i-\alpha_i)x} - 1)\right] = 0,$$

it follows that

$$\frac{\partial f_i}{\partial g_i} = \frac{\alpha_i}{\alpha_i - \delta_i - \pi_i}(1 - e^{-(\alpha_i-\delta_i-\pi_i)x}). \tag{4.3.6}$$

By following the same approach, we have

$$A_i(x) = e^{\beta_i x} g_i + \frac{\delta_i}{\beta_i}(1 - e^{\beta_i x}) - \frac{\pi_i f_{i+1}^2}{2\alpha_{i+1} - \beta_i}(e^{(2\alpha_{i+1}-\beta_i)x} - 1)$$

$$+2\alpha_{i+1}\pi_i f_{i+1}e^{\beta_i x}\int_0^x e^{(2\alpha_{i+1}-\beta_i)s}\int_0^s e^{\alpha_{i+1}\tau}A_{i+1}(\tau)d\tau ds$$
$$-\pi_i\alpha_{i+1}^2 e^{\beta_i x}\int_0^x e^{(2\alpha_{i+1}-\beta_i)s}\left(\int_0^s e^{-\alpha_{i+1}\tau}A_{i+1(\tau)}d\tau\right)^2 ds,$$

then

$$\frac{\partial A_i}{\partial f_{i+1}} = 0 = \frac{\partial g_i}{\partial f_{i+1}} - 2\pi_i f_{i+1}\int_0^x e^{(2\alpha_{i+1}-\beta_i)s}ds + 2\pi_i\alpha_{i+1}\int_0^x e^{(2\alpha_{i+1}-\beta_i)s}\left[\int_0^s e^{-\alpha_{i+1}\tau}A_{i+1}(\tau)d\tau\right]ds.$$

As a results,

$$\left.\frac{\partial g_i}{\partial f_{i+1}}\right|_{\boldsymbol{S}=(1,1,\cdots,1)} = \frac{2\pi_i}{\delta_i+\pi_i-\alpha_{i+1}}(1-e^{-(\delta_i+\pi_i-\alpha_{i+1})x}). \tag{4.3.7}$$

Hence,

$$\mathbf{E}\left(n_i^{(Q)}\right) = 2^i n_0^{(Q)}e^{(\alpha_0-\alpha_i)x}\prod_{j=0}^{i-1}\frac{\alpha_j\pi_j}{(\alpha_j-\delta_j-\pi_j)(\delta_j+\pi_j-\alpha_{j+1})}(1-e^{-(\alpha_j-\delta_j-\pi_j)x})(1-e^{-(\delta_j+\pi_j-\alpha_{j+1})x}). \tag{4.3.8}$$

To ease reading, let us set:

$$\Upsilon_j(x) = \frac{\alpha_j\pi_j}{(\alpha_j-\delta_j-\pi_j)(\delta_j+\pi_j-\alpha_{j+1})}(1-e^{-(\alpha_j-\delta_j-\pi_j)x})(1-e^{-(\delta_j+\pi_j-\alpha_{j+1})x}), \tag{4.3.9}$$

then it follows

$$\mathbf{E}\left(n_i^{(Q)}\right) = 2^i n_0^{(Q)}e^{(\alpha_0-\alpha_i)x}\prod_{j=0}^{i-1}\Upsilon_j(x). \tag{4.3.10}$$

We also have

$$\begin{aligned}
\mathbf{E}\left(n_i^{(A)}\right) &= \left.\frac{\partial G(\boldsymbol{S},t)}{\partial A_i}\right|_{\boldsymbol{S}=(1,1,\cdots,1)}\\
&= \left.\frac{\partial Q_0^{n_0^{(Q)}}}{\partial A_i}\right|_{\boldsymbol{S}=(1,1,\cdots,1)}\\
&= n_0^{(Q)}Q_0^{n_0^{(Q)}-1}\left.\frac{\partial Q_0}{\partial A_i}\right|_{\boldsymbol{S}=(1,1,\cdots,1)}\\
&= n_0^{(Q)}Q_0^{n_0^{(Q)}-1}\frac{\partial Q_0}{\partial f_0}\left(\prod_{j=0}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\frac{\partial f_i}{\partial g_i}\left.\frac{\partial g_i}{\partial A_i}\right|_{\boldsymbol{S}=(1,1,\cdots,1)},
\end{aligned}$$

$$= \mathbf{E}\left(n_i^{(Q)}\right)\frac{\partial f_i}{\partial g_i}\frac{\partial g_i}{\partial A_i}\bigg|_{\boldsymbol{S}=(1,1,\cdots,1)} \tag{4.3.11}$$

Yet,

$$\frac{\partial g_i}{\partial A_i} = e^{-(\delta_i+\pi_i)x},$$

from (4.3.6) and (4.3.11), it follows

$$\mathbf{E}\left(n_i^{(A)}\right) = \frac{2^i n_0^{(Q)}\alpha_i e^{(\alpha_0-\alpha_i-\delta_i-\pi_i)x}(1-e^{-(\alpha_i-\delta_i-\pi_i)x})}{\alpha_i-\pi_i-\delta_i}\prod_{j=0}^{i-1}\Upsilon_j(x)$$

4.3.2 Variance of cell counts

To compute the variances of quiescent and activated cells of each generation, we refer to (3.2.6)

$$\mathbf{Var}\ (n_i^{(Q)}) = \mathbf{E}\ [n_i^{(Q)}(n_i^{(Q)}-1)] + \mathbf{E}\ (n_i^{(Q)}) - \mathbf{E}^{\ 2}(n_i^{(Q)}) \tag{4.3.12}$$

$$\mathbf{Var}\ (n_i^{(A)}) = \mathbf{E}\ [n_i^{(A)}(n_i^{(A)}-1)] + \mathbf{E}\ (n_i^{(A)}) - \mathbf{E}^{\ 2}(n_i^{(A)}). \tag{4.3.13}$$

From the above formulas, it follows that the calculation of each variance is reduced to the evaluation of the corresponding second moments terms. That is to say

$$\begin{cases} \mathbf{E}\ [n_i^{(Q)}(n_i^{(Q)}-1)] = \dfrac{\partial^2 G(\boldsymbol{S},t)}{\partial Q_i^2}\bigg|_{\boldsymbol{S}=(1,1,\cdots,1)} \\ \\ \mathbf{E}\ [n_i^{(A)}(n_i^{(A)}-1)] = \dfrac{\partial^2 G(\boldsymbol{S},t)}{\partial A_i^2}\bigg|_{\boldsymbol{S}=(1,1,\cdots,1)}. \end{cases} \tag{4.3.14}$$

We have

$$\begin{aligned} \mathbf{E}\ [n_i^{(Q)}(n_i^{(Q)}-1)] &= \frac{\partial^2 G(\boldsymbol{S},t)}{\partial Q_i^2}\bigg|_{\boldsymbol{S}=(1,1,\cdots,1)} \\ &= \frac{\partial}{\partial Q_i}\left(n_0^{(Q)}Q_0^{n_0^{(Q)}-1}\frac{\partial Q a_0}{\partial Q_i}\right)\bigg|_{\boldsymbol{S}=(1,1,\cdots,1)} \\ &= n_0^{(Q)}\left[(n_0^{(Q)}-1)Q_0^{n_0^{(Q)}-2}\left(\frac{\partial Q_0}{\partial Q_i}\right)^2 + Q_0^{n_0^{(Q)}-1}\frac{\partial^2 Q_0}{\partial Q_i^2}\right]\bigg|_{\boldsymbol{S}=(1,1,\cdots,1)} \end{aligned} \tag{4.3.15}$$

The challenge here is to calculate $\frac{\partial^2 Q_0}{\partial Q_i^2}$, given that the other terms appearing in (4.3.15) are known.

By using the result from (4.3.3)

$$\begin{aligned}
\frac{\partial^2 Q_0}{\partial Q_i^2} &= \frac{\partial}{\partial Q_i}\left[\frac{\partial Q_0}{\partial f_0}\left(\prod_{j=0}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\frac{\partial f_i}{\partial Q_i}\right] \\
&= \frac{\partial^2 Q_0}{\partial Q_i \partial f_0}\left(\prod_{j=0}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\frac{\partial f_i}{\partial Q_i} + \frac{\partial Q_0}{\partial f_0}\left(\prod_{j=0}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\frac{\partial^2 f_i}{\partial Q_i^2} \\
&\quad + \frac{\partial Q_0}{\partial f_0}\frac{\partial f_i}{\partial Q_i}\left[\sum_{k=0}^{i-1}\frac{\partial^2 f_k}{\partial Q_i \partial g_k}\left(\prod_{\substack{j=0\\ j\neq k}}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\frac{\partial g_k}{\partial f_{k+1}} + \sum_{k=0}^{i-1}\frac{\partial^2 g_k}{\partial Q_i \partial f_{k+1}}\left(\prod_{\substack{j=0\\ j\neq k}}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\frac{\partial f_k}{\partial g_k}\right]
\end{aligned} \tag{4.3.16}$$

Now let us compute all the unknown terms appearing in (4.3.16) that is to say:

$$\frac{\partial^2 Q_0}{\partial Q_i \partial f_0}, \quad \frac{\partial^2 f_i}{\partial Q_i^2}, \quad \frac{\partial^2 f_k}{\partial Q_i \partial g_k} \quad \text{and} \quad \frac{\partial^2 g_k}{\partial Q_i \partial f_{k+1}}$$

We have

$$\begin{aligned}
\frac{\partial^2 Q_0}{\partial Q_i \partial f_0} &= \frac{\partial}{\partial Q_i}\left(\frac{\partial Q_0}{\partial f_0}\right) \\
&= \frac{\partial^2 Q_0}{\partial f_0^2}\left(\prod_{j=0}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\frac{\partial f_i}{\partial Q_i}.
\end{aligned}$$

Since

$$\frac{\partial^2 Q_0}{\partial f_0^2} = 0,$$

it follows that

$$\frac{\partial^2 Q_0}{\partial Q_i \partial f_0} = 0. \tag{4.3.17}$$

We also have

$$\frac{\partial^2 f_i}{\partial Q_i^2} = \frac{\partial}{\partial Q_i}\left(e^{-\alpha_i x}\right) = 0,$$

$$\frac{\partial^2 f_k}{\partial Q_i \partial g_k} = \frac{\partial}{\partial Q_i}\left[\frac{\alpha_k}{\alpha_k - \delta_k - \pi_k}\left(1 - e^{(\alpha_k - \delta_k - \pi_k)x}\right)\right] = 0, \tag{4.3.18}$$

To compute $\frac{\partial^2 g_k}{\partial Q_i \partial f_{k+1}}$, we have:

$$\frac{\partial^2 g_k}{\partial Q_i \partial f_{k+1}} = \frac{\partial}{\partial Q_i}\left(\frac{\partial g_k}{\partial f_{k+1}}\right) \tag{4.3.19}$$

Let

$$h_k = \frac{\partial g_k}{\partial f_{k+1}},$$

then (4.3.19) becomes

$$\begin{aligned}
\frac{\partial^2 g_k}{\partial Q_i \partial f_{k+1}} &= \frac{\partial h_k}{\partial Q_i} \\
&= \frac{\partial h_k}{\partial f_l}\left(\prod_{j=l}^{i-1} \frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\frac{\partial f_i}{\partial Q_i} \\
&= \frac{\partial^2 g_k}{\partial f_l \partial f_{k+1}}\left(\prod_{j=l}^{i-1} \frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\frac{\partial f_i}{\partial Q_i}. \qquad (4.3.20)
\end{aligned}$$

Let us now compute $\frac{\partial^2 g_k}{\partial f_l \partial f_{k+1}}$.

We have

$$\frac{\partial^2 g_k}{\partial f_l \partial f_{k+1}} = \frac{\partial}{\partial f_l}\left(\frac{\partial g_k}{\partial f_{k+1}}\right)$$

According to the expression of $\frac{\partial g_k}{\partial f_{k+1}}$ computed earlier, it appears that for $l \neq k+1$,

$$\frac{\partial^2 g_k}{\partial f_l \partial f_{k+1}} = 0,$$

and

$$\frac{\partial^2 g_k}{\partial f_{k+1}^2} = \frac{2\pi_k}{2\alpha_{k+1} - \beta_k}\left(e^{(2\alpha_{k+1}-\beta_k)x} - 1\right) = \frac{2\pi_k}{2\alpha_{k+1} - \delta_k - \pi_k}\left(e^{(2\alpha_{k+1}-\delta_k-\pi_k)x} - 1\right)$$

hence

$$\frac{\partial^2 g_k}{\partial Q_i \partial f_{k+1}} = \frac{\partial^2 g_k}{\partial f_{k+1}^2}\left(\prod_{j=k+1}^{i-1} \frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\frac{\partial f_i}{\partial Q_i}. \tag{4.3.21}$$

From (4.3.16), it follows that

$$\frac{\partial^2 Q_0}{\partial Q_i^2} = \frac{\partial Q_0}{\partial f_0}\left(\frac{\partial f_i}{\partial Q_i}\right)^2 \sum_{k=0}^{i-1} \frac{\partial^2 g_k}{\partial f_{k+1}^2}\left(\prod_{j=k+1}^{i-1} \frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\left(\prod_{\substack{j=0\\ j\neq k}}^{i-1} \frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\frac{\partial f_k}{\partial g_k}, \tag{4.3.22}$$

replacing (4.3.22) into (4.3.15) one obtains

$$\begin{aligned}
\mathbf{E}\left[n_i^{(Q)}(n_i^{(Q)}-1)\right] &= n_0^{(Q)}\Bigg\{(n_0^{(Q)}-1)Q_0^{n_0^{(Q)}-2}\left(\frac{\partial Q_0}{\partial f_0}\right)^2\left(\frac{\partial f_i}{\partial Q_i}\right)^2\left(\prod_{j=0}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)^2 \\
&\quad + Q_0^{n_0^{(Q)}}\frac{\partial Q_0}{\partial f_0}\left(\frac{\partial f_i}{\partial Q_i}\right)^2\sum_{k=0}^{i-1}\left[\frac{\partial^2 g_k}{\partial f_{k+1}^2}\left(\prod_{j=k+1}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\right. \\
&\quad \left.\left(\prod_{\substack{j=0\\ j\neq k}}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\frac{\partial f_k}{\partial g_k}\right]\Bigg\}\Bigg|_{\boldsymbol{S}=(1,1,\cdots,1)} \qquad (4.3.23)\\
&= n_0^{(Q)}\Bigg((n_0^{(Q)}-1)e^{2(\alpha_0-\alpha_i)x}\left(\prod_{j=0}^{i-1}4\big(\Upsilon_j(x)\big)^2\right) \\
&\quad + e^{(\alpha_0-2\alpha_i)x}\sum_{k=0}^{i-1}\Bigg[\frac{2\pi_k}{2\alpha_{k+1}-\delta_k-\pi_k}\left(e^{(2\alpha_{k+1}-\delta_k-\pi_k)x}-1\right)\left(\prod_{j=k+1}^{i-1}2\Upsilon_j(x)\right) \\
&\quad \left(\prod_{\substack{j=0\\ j\neq k}}^{i-1}2\Upsilon_j(x)\right)\frac{\alpha_k}{\alpha_k-\delta_k-\pi_k}\left(1-e^{-(\alpha_k-\delta_k-\pi_k)x}\right)\Bigg]\Bigg),
\end{aligned}$$

(4.3.24)

replacing (4.3.8) and (4.3.24) into (4.3.12), we end with:

$$\begin{aligned}
\mathbf{Var}\left(n_i^{(Q)}\right) &= n_0^{(Q)}\Bigg((n_0^{(Q)}-1)e^{2(\alpha_0-\alpha_i)x}\left(\prod_{j=0}^{i-1}4\big(\Upsilon_j(x)\big)^2\right) \\
&\quad + e^{(\alpha_0-2\alpha_i)x}\sum_{k=0}^{i-1}\Bigg[\frac{2\pi_k}{2\alpha_{k+1}-\delta_k-\pi_k}\left(e^{(2\alpha_{k+1}-\delta_k-\pi_k)x}-1\right)\left(\prod_{j=k+1}^{i-1}2\Upsilon_j(x)\right)
\end{aligned}$$

$$\left(\prod_{\substack{j=0\\j\neq k}}^{i-1} 2\Upsilon_j(x)\right) \frac{\alpha_k}{\alpha_k-\delta_k-\pi_k}(1-e^{-(\alpha_k-\delta_k-\pi_k)x})\Bigg]\Bigg) + 2^i n_0^{(Q)} e^{(\alpha_0-\alpha_i)x} \prod_{j=0}^{i-1}\Upsilon_j(x)$$
$$-4^i \left(n_0^{(Q)}\right)^2 e^{2(\alpha_0-\alpha_i)x} \prod_{j=0}^{i-1} \left(\Upsilon_j(x)\right)^2$$

Using the above reasoning, we also compute **Var** $(n_i^{(Q)})$.

As we mentioned in (4.3.14), the challenge is to compute the second moment. We therefore have:

$$\begin{aligned}
\mathbf{E}\left[n_i^{(A)}(n_i^{(A)}-1)\right] &= \left.\frac{\partial^2 G(\boldsymbol{S},t)}{\partial A_i^2}\right|_{\boldsymbol{S}=(1,1,\cdots,1)} \\
&= \frac{\partial}{\partial A_i}\left(n_0^{(Q)} Q_0^{n_0^{(Q)}-1}\frac{\partial Q_0}{\partial A_i}\right)\Bigg|_{\boldsymbol{S}=(1,1,\cdots,1)} \\
&= n_0^{(Q)}\left[(n_0^{(Q)}-1)Q_0^{n_0^{(Q)}-2}\left(\frac{\partial Q_0}{\partial A_i}\right)^2 + Q_0^{n_0^{(Q)}-1}\frac{\partial^2 Q_0}{\partial A_i^2}\right]\Bigg|_{\boldsymbol{S}=(1,1,\cdots,1)}
\end{aligned} \tag{4.3.25}$$

All the terms in (4.3.25) except $\frac{\partial^2 Q_0}{\partial Q_i^2}$ are known, hence, let us compute the unknown term.

$$\begin{aligned}
\frac{\partial^2 Q_0}{\partial A_i^2} &= \frac{\partial}{\partial A_i}\left[\frac{\partial Q_0}{\partial f_0}\left(\prod_{j=0}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\frac{\partial f_i}{\partial g_i}\frac{\partial g_i}{\partial A_i}\right] \\
&= \frac{\partial^2 Q_0}{\partial A_i \partial f_0}\left(\prod_{j=0}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\frac{\partial f_i}{\partial g_i}\frac{\partial g_i}{\partial A_i} + \frac{\partial Q_0}{\partial f_0}\frac{\partial^2 f_i}{\partial A_i \partial g_i}\frac{\partial g_i}{\partial A_i}\left(\prod_{j=0}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right) \\
&\quad + \frac{\partial Q_0}{\partial f_0}\frac{\partial f_i}{\partial g_i}\frac{\partial^2 g_i}{\partial A_i^2}\left(\prod_{j=0}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right) \\
&\quad + \frac{\partial Q_0}{\partial f_0}\frac{\partial f_i}{\partial g_i}\frac{\partial g_i}{\partial A_i}\left[\sum_{k=0}^{i-1}\frac{\partial^2 f_k}{\partial A_i \partial g_k}\left(\prod_{\substack{j=0\\j\neq k}}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\frac{\partial g_k}{\partial f_{k+1}} + \sum_{k=0}^{i-1}\frac{\partial^2 g_k}{\partial A_i \partial f_{k+1}}\left(\prod_{\substack{j=0\\j\neq k}}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\frac{\partial f_k}{\partial g_k}\right].
\end{aligned} \tag{4.3.26}$$

As before, we have to compute the following terms:

$$\frac{\partial^2 Q_0}{\partial A_i \partial f_0}, \qquad \frac{\partial^2 f_i}{\partial A_i \partial g_i}, \qquad \frac{\partial^2 g_i}{\partial A_i^2}, \qquad \frac{\partial^2 f_k}{\partial A_i \partial g_k} \quad \text{and} \quad \frac{\partial^2 g_k}{\partial A_i \partial f_{k+1}}.$$

Similarly to the previous calculations, the four first terms vanish, and the evaluation of the last term is straight forward and we have:

$$\frac{\partial^2 g_k}{\partial A_i \partial f_{k+1}} = \frac{\partial^2 g_k}{\partial f_{k+1}^2}\left(\prod_{j=k+1}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\frac{\partial f_i}{\partial g_i}\frac{\partial g_i}{\partial A_i} \tag{4.3.27}$$

with

$$\frac{\partial^2 g_k}{\partial f_{k+1}^2} = \frac{2\pi_k}{2\alpha_{k+1}-\beta_k}\left(e^{(2\alpha_{k+1}-\beta_k)x}-1\right) = \frac{2\pi_k}{2\alpha_{k+1}-\delta_k-\pi_k}\left(e^{(2\alpha_{k+1}-\delta_k-\pi_k)x}-1\right)$$

hence,

$$\frac{\partial^2 Q_0}{\partial A_i^2} = \frac{\partial Q_0}{\partial f_0}\left(\frac{\partial f_i}{\partial g_i}\right)^2\left(\frac{\partial g_i}{\partial A_i}\right)^2\sum_{k=0}^{i-1}\left[\frac{\partial^2 g_k}{\partial f_{k+1}^2}\left(\prod_{j=k+1}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\left(\prod_{\substack{j=0\\ j\neq k}}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\frac{\partial f_k}{\partial g_k}\right].$$

From (4.3.28) into (4.3.25) it follows

$$\begin{aligned}
\mathbf{E}\left[n_i^{(A)}(n_i^{(A)}-1)\right] &= n_0^{(Q)}\Bigg\{(n_0^{(Q)}-1)Q_0^{n_0^{(Q)}-2}\left(\frac{\partial Q_0}{\partial f_0}\right)^2\left(\frac{\partial f_i}{\partial g_i}\right)^2\left(\frac{\partial g_i}{\partial A_i}\right)^2\left(\prod_{j=0}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)^2\\
&\quad + Q_0^{n_0^{(Q)}}\frac{\partial Q_0}{\partial f_0}\left(\frac{\partial f_i}{\partial g_i}\right)^2\left(\frac{\partial g_i}{\partial A_i}\right)^2\sum_{k=0}^{i-1}\left[\frac{\partial^2 g_k}{\partial f_{k+1}^2}\right.\\
&\quad \left.\left(\prod_{j=k+1}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\left(\prod_{\substack{j=0\\ j\neq k}}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\frac{\partial f_k}{\partial g_k}\right]\Bigg\}\Bigg|_{\boldsymbol{S}=(1,1,\cdots,1)} \qquad (4.3.28)\\
&= n_0^{(Q)}\Bigg((n_0^{(Q)}-1)e^{2(\alpha_0-\alpha_i)x}\left(\prod_{j=0}^{i-1}\left(\Upsilon_j(x)\right)^2\right)\\
&\quad + e^{(\alpha_0-2\delta_i-2\pi_i)x}\frac{\alpha_i^2}{(\alpha_i-\delta_i-\pi_i)^2}(1-e^{-(\alpha_i-\delta_i-\pi_i)x})^2
\end{aligned}$$

$$\sum_{k=0}^{i-1}\left[\frac{2\pi_k}{2\alpha_{k+1}-\delta_k-\pi_k}\left(e^{(2\alpha_{k+1}-\delta_k-\pi_k)x}-1\right)\left(\prod_{j=k+1}^{i-1}2\Upsilon_j(x)\right)\quad\left(\prod_{\substack{j=0\\j\neq k}}^{i-1}2\Upsilon_j(x)\right)\right.$$
$$\left.\left.\frac{\alpha_k}{\alpha_k-\delta_k-\pi_k}(1-e^{-(\alpha_k-\delta_k-\pi_k)x})\right]\right),\tag{4.3.29}$$

hence

$$\begin{aligned}\textbf{Var}\ (n_i^{(A)}) &= n_0^{(Q)}\left((n_0^{(Q)}-1)e^{2(\alpha_0-\alpha_i)x}\left(\prod_{j=0}^{i-1}\left(\Upsilon_j(x)\right)^2\right)\right.\\ &\quad + e^{(\alpha_0-2\delta_i-2\pi_i)x}\frac{\alpha_i^2}{(\alpha_i-\delta_i-\pi_i)^2}(1-e^{-(\alpha_i-\delta_i-\pi_i)x})^2\\ &\quad \sum_{k=0}^{i-1}\left[\frac{2\pi_k}{2\alpha_{k+1}-\delta_k-\pi_k}\left(e^{(2\alpha_{k+1}-\delta_k-\pi_k)x}-1\right)\left(\prod_{j=k+1}^{i-1}2\Upsilon_j(x)\right)\quad\left(\prod_{\substack{j=0\\j\neq k}}^{i-1}2\Upsilon_j(x)\right)\right.\\ &\quad \left.\left.\frac{\alpha_k}{\alpha_k-\delta_k-\pi_k}(1-e^{-(\alpha_k-\delta_k-\pi_k)x})\right]\right)+\frac{2^i n_0^{(Q)}\alpha_i e^{(\alpha_0-\alpha_i-\delta_i-\pi_i)x}(1-e^{-(\alpha_i-\delta_i-\pi_i)x})}{\alpha_i-\pi_i-\delta_i}\times\\ &\quad \left(\prod_{j=0}^{i-1}\Upsilon_j(x)\right)-\frac{4^i\left(n_0^{(Q)}\right)^2\alpha_i^2 e^{2(\alpha_0-\alpha_i-\delta_i-\pi_i)x}(1-e^{-(\alpha_i-\delta_i-\pi_i)x})^2}{(\alpha_i-\pi_i-\delta_i)^2}\left(\prod_{j=0}^{i-1}\left(\Upsilon_j(x)\right)^2\right).\end{aligned}\tag{4.3.30}$$

4.3.3 Covariance of cell counts

To be able to analyze the available data, it is necessary to calculate the variances of the sum of the quiescent cells and the active cells. This raises the question of covariance. Indeed, we have

$$\textbf{Var}\ (n_i^{(Q)}+n_i^{(A)}) = \textbf{Var}\ (n_i^{(Q)})+\textbf{Var}\ (n_i^{(A)})+2\times\textbf{Cov}\ (n_i^{(Q)},n_i^{(A)})\tag{4.3.31}$$

Given that we already know what are the two variances, the only challenge here will be to compute the covariance. The covariance of is given by

$$\textbf{Cov}\ (n_i^{(Q)},n_i^{(A)}) = \textbf{E}\ (n_i^{(Q)}n_i^{(A)})-\textbf{E}\ (n_i^{(Q)})\textbf{E}\ (n_i^{(A)}),\tag{4.3.32}$$

and

$$
\begin{aligned}
\mathbf{E}\ (n_i^{(Q)} n_i^{(A)}) &= \left.\frac{\partial^2 G(\boldsymbol{S},t)}{\partial Q_i \partial A_i}\right|_{\boldsymbol{S}=(1,\cdots,1)} \\
&= \left.\frac{\partial}{\partial Q_i}\left(\frac{\partial Q_0^{n_0^{(Q)}}}{\partial A_i}\right)\right|_{\boldsymbol{S}=(1,\cdots,1)} \\
&= \left. n_0^{(Q)}\left[(n_0^{(Q)}-1) Q_0^{n_0^{(Q)}-2}\frac{\partial Q_0}{\partial Q_i}\frac{\partial Q_0}{\partial A_i} + Q_0^{n_0^{(Q)}-1}\frac{\partial^2 Q_0}{\partial Q_i \partial A_i}\right]\right|_{\boldsymbol{S}=(1,\cdots,1)}
\end{aligned}
\tag{4.3.33}
$$

In (4.3.33), the only unknown term is $\frac{\partial^2 Q_0}{\partial Q_i \partial A_i}$, which by using the same approach as in the case of variances can be computed as follow

$$
\begin{aligned}
\frac{\partial^2 Q_0}{\partial Q_i \partial A_i} &= \frac{\partial}{\partial Q_i}\left[\frac{\partial Q_0}{\partial f_0}\left(\prod_{j=0}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\frac{\partial f_i}{\partial g_i}\frac{\partial g_i}{\partial A_i}\right] \\
&= \frac{\partial^2 Q_0}{\partial Q_i \partial f_0}\left(\prod_{j=0}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\frac{\partial f_i}{\partial g_i}\frac{\partial g_i}{\partial A_i} + \frac{\partial Q_0}{\partial f_0}\frac{\partial^2 f_i}{\partial Q_i \partial g_i}\frac{\partial g_i}{\partial A_i}\left(\prod_{j=0}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right) \\
&\quad + \frac{\partial Q_0}{\partial f_0}\frac{\partial f_i}{\partial g_i}\frac{\partial^2 g_i}{\partial Q_i \partial A_i}\left(\prod_{j=0}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right) \\
&\quad + \frac{\partial Q_0}{\partial f_0}\frac{\partial f_i}{\partial g_i}\frac{\partial g_i}{\partial A_i}\left[\sum_{k=0}^{i-1}\frac{\partial^2 f_k}{\partial Q_i \partial g_k}\left(\prod_{\substack{j=0\\ j\neq k}}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\frac{\partial g_k}{\partial f_{k+1}}\right. \\
&\quad \left. + \sum_{k=0}^{i-1}\frac{\partial^2 g_k}{\partial Q_i \partial f_{k+1}}\left(\prod_{\substack{j=0\\ j\neq k}}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\frac{\partial f_k}{\partial g_k}\right].
\end{aligned}
\tag{4.3.34}
$$

As noted in our previous calculations,

$$\begin{cases} \frac{\partial^2 Q_0}{\partial Q_i \partial f_0} = 0 \\ \frac{\partial^2 f_i}{\partial Q_i \partial g_i} = 0 \\ \frac{\partial^2 g_i}{\partial Q_i \partial A_i} = 0 \\ \frac{\partial^2 f_k}{\partial Q_i \partial g_k} = 0 \\ \frac{\partial^2 g_k}{\partial Q_i \partial f_{k+1}} \neq 0, \end{cases} \tag{4.3.35}$$

it follows that

$$\frac{\partial^2 Q_0}{\partial Q_i \partial A_i} = \frac{\partial Q_0}{\partial f_0}\frac{\partial f_i}{\partial g_i}\frac{\partial g_i}{\partial A_i}\sum_{k=0}^{i-1}\frac{\partial^2 g_k}{\partial Q_i \partial f_{k+1}}\left(\prod_{\substack{j=0\\ j\neq k}}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\frac{\partial f_k}{\partial g_k}, \tag{4.3.36}$$

therefore

$$\begin{aligned}
\mathbf{E}\,(n_i^{(Q)}n_i^{(A)}) &= n_0^{(Q)}\Bigg[\left(n_0^{(Q)}-1\right)\frac{\partial Q_0}{\partial Q_i}\frac{\partial Q_0}{\partial A_i} + \frac{\partial Q_0}{\partial f_0}\frac{\partial f_i}{\partial g_i}\frac{\partial g_i}{\partial A_i}\sum_{k=0}^{i-1}\frac{\partial^2 g_k}{\partial Q_i \partial f_{k+1}}\left(\prod_{\substack{j=0\\ j\neq k}}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\frac{\partial f_k}{\partial g_k}\Bigg]\\
&= n_0^{(Q)}\Bigg[\left(n_0^{(Q)}-1\right)\left(\frac{\partial Q_0}{\partial f_0}\right)^2\left(\prod_{j=0}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)^2\frac{\partial f_i}{\partial Q_i}\frac{\partial f_i}{\partial g_i}\frac{\partial g_i}{\partial A_i}\\
&\quad +\frac{\partial Q_0}{\partial f_0}\frac{\partial f_i}{\partial g_i}\frac{\partial g_i}{\partial A_i}\sum_{k=0}^{i-1}\frac{\partial^2 g_k}{\partial f_{k+1}^2}\left(\prod_{j=k+1}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\frac{\partial f_i}{\partial Q_i}\left(\prod_{\substack{j=0\\ j\neq k}}^{i-1}\frac{\partial f_j}{\partial g_j}\frac{\partial g_j}{\partial f_{j+1}}\right)\frac{\partial f_k}{\partial g_k}\Bigg]\\
&= n_0^{(Q)}\Bigg[\frac{4^i\alpha_i(n_0^{(Q)}-1)}{\alpha_i-\delta_i-\pi_i}e^{(2\alpha_0-\alpha_i-\delta_i-\pi_i)x}\left(1-e^{-(\alpha_i-\delta_i-\pi_i)x}\right)\prod_{j=0}^{i-1}\Upsilon_j^2(x)\\
&\quad +2^{2i-1}e^{-\alpha_i x}\sum_{k=0}^{i-1}\frac{2^{-k}\alpha_k\pi_k}{(2\alpha_{k+1}-\delta_k-\pi_k)(\alpha_k-\delta_k-\pi_k)}\left(\prod_{j=k+1}^{i-1}\Upsilon_j(x)\right)\left(\prod_{\substack{j=0\\ j\neq k}}^{i-1}\Upsilon_j(x)\right)\\
&\quad \left(e^{(2\alpha_{k+1}-\delta_k-\pi_k)x}-1\right)\left(1-e^{-(\alpha_k-\delta_k-\pi_k)x}\right)\Bigg]
\end{aligned} \tag{4.3.37}$$

Thus

$$\begin{aligned}\textbf{Cov}\ (n_i^{(Q)}, n_i^{(A)}) \quad = \quad & n_0^{(Q)}\Bigg[\frac{4^i\alpha_i(n_0^{(Q)}-1)}{\alpha_i-\delta_i-\pi_i}e^{(2\alpha_0-\alpha_i-\delta_i-\pi_i)x}\big(1-e^{-(\alpha_i-\delta_i-\pi_i)x}\big)\prod_{j=0}^{i-1}\Upsilon_j^2(x) \\ & +2^{2i-1}e^{-\alpha_i x}\sum_{k=0}^{i-1}\frac{2^{-k}\alpha_k\pi_k}{(2\alpha_{k+1}-\delta_k-\pi_k)(\alpha_k-\delta_k-\pi_k)}\left(\prod_{j=k+1}^{i-1}\Upsilon_j(x)\right)\left(\prod_{\substack{j=0\\ j\neq k}}^{i-1}\Upsilon_j(x)\right) \\ & \big(e^{(2\alpha_{k+1}-\delta_k-\pi_k)x}-1\big)\big(1-e^{-(\alpha_k-\delta_k-\pi_k)x}\big)\Bigg]-\frac{4^i\alpha_i\big(n_0^{(Q)}\big)^2}{\alpha_i-\delta_i-\pi_i}\left(\prod_{j=k+1}^{i-1}\Upsilon_j^2(x)\right) \\ & \times e^{(2\alpha_0-2\alpha_i-\delta_i-\pi_i)x}\big(1-e^{-(\alpha_i-\delta_i-pi_i)x}\big).\end{aligned} \tag{4.3.38}$$

5. Scene for fitting models to data

So far we built a mathematical model of T cell dynamics that agrees with the information from T cell biology. The question is how close the built model is from the experimental data? Furthermore how can we estimate the values of the parameters of the model? The approach that we used here is essentially to fit the model to data by using the maximum of likelihood approach. We presented a brief description of the MLE approach in the chapter 3. Let us now give the step for its implementation.

5.1 Maximum Likelihood Algorithm

We present a simple form of the algorithm of Maximum likelihood according to [37].

The algorithm can be split in two parts. The first part consists to compute through a function called FUN the log-likelihood. The function takes two inputs (the parameter value θ and the data X) and return as output the corresponding log-likelihood as illustrated below:

$$FUN(\theta,\, X)\text{: Return: } \ln\Big(L(\theta, X)\Big).$$

The second part of the algorithm basically invokes the function FUN several times. Each time a different guess of $\hat{\theta}$ is provided as an input of the function FUN which returns the corresponding likelihood and store it until the algorithm find a "**good guess**" according to a pre-specified criterion. That guess is the approximate solution of (3.3.8). Here are the steps:

Step 1: Make a guess θ_0

Step 2: Call FUN to compute $l_0 = FUN(\theta_0, X)$

Step 3: Make another guess θ_i

Step 4: Call FUN to compute $l_i = FUN(\theta_i, X)$

Step 5: If $l_i > l_0$, then re-define $\theta_0 \leftarrow \theta_i$ and $l_0 \leftarrow l_i$

Step 6: If θ_0 is a good guess, then end the program. Otherwise, return to **Step 3**

The **Step 6** is a bit tricky and there is a whole field studying how this step can performed and the challenges. In this work, instead of looking in that direction, we rather build a framework to take advantage of R programming and his predefined functions such as **Optim** to perform this task. Even though we know how to implement the MLE algorithm, there is still a big challenge on the distribution to describe the process. How do we chose the probability distribution to describe the proliferation process?

5.2 Choice of the probability distribution

The choice of a distribution that best describes the disposable data is a big challenge, especially when we do not have historical data or when the data are skewed [38], The guideline by Damodaran [39] give us some tips that help to choose the right distribution to approach available data. The choice of the distribution relies on the following questions:

- Is our data discrete or continuous?
- Can we estimate the outcomes and the probability?
- Is our data symmetric or asymmetric?
- Are the outlier positive or negative?
- What about the dispersion? Is the variance bigger than the mean?

For the last point, our expectation is that the variance of the distribution should be greater than the mean. By looking at the data from CFSE labeling, it guides us and comforts that our expectation holds. Regarding the other points, we just follow the guidelines for choosing a distribution from Damodaran [39] as it is described by the diagram (see figure (5.1)).

By following the chart, it results that the right distribution to be chosen is the **negative binomial**. We have the required distribution and the method to fit the model to data from CFSE labeling, It remains to compute the parameters of the chosen distribution. We also know that the negative binomial distribution can be approached by the Poisson-Gamma mixture distribution. The reason is that the negative binomial can describe as over-dispersed Poisson process. Indeed let r been

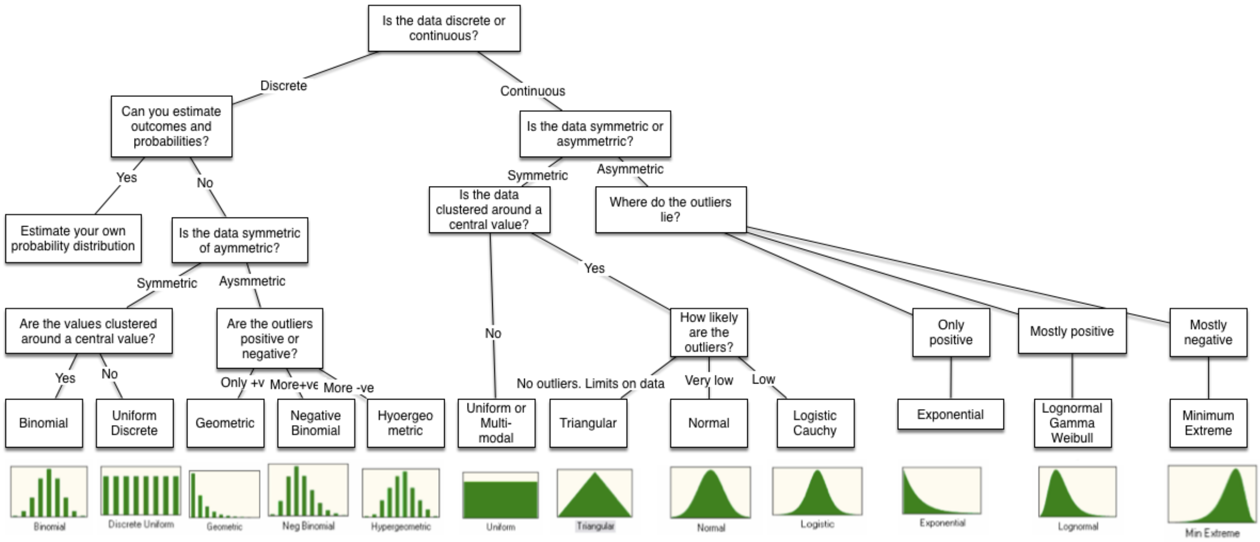

Figure 5.1: Chart to choose a distribution (Image from https://www.valuewalk.com/wp-content/uploads/2016/05/DCF-Myth-8.png)

the dispersion parameter from Poisson with parameter λ, we have

$$Poisson(\lambda) = \lim_{r\to\infty} NB\Big(r, \frac{\lambda}{\lambda + r}\Big), \tag{5.2.1}$$

furthermore, we have

$$\binom{k+r-1}{k} = \frac{\Gamma(k+r)}{k!\Gamma(k)} \tag{5.2.2}$$

For reasons of analysis and programming, we will use the Poisson-gamma mixture parametrization of the negative binomial. The following parametrization is given by

$$P(X = x) = \frac{\Gamma(x + 1/b)}{x!\Gamma(1/b)}(mb)^x(1 + mb)^{-x+1/b}, \tag{5.2.3}$$

Where

$$m = \mathbf{E}\ (x), \quad b = \text{dispersion parameter} \quad \text{and} \quad \mathbf{Var}\ (x) = m(1 + mb).$$

5.2.1 The data

The available data for model fitting are originated from the paper by Liu et al. [40], describing the total number of CD4 and CD8 T cell division for seven observation time points and for each generation up to the 6th generation. In the paper, data are used for the determination of threshold for cellular division. Hence, presents forty-nine observation. It is important to note that the method used by Liu et al. aligns with the direction we are taking in this work as they use CFSE labeling and statistical analysis to determine the division threshold. Even though one can argue on the different steps of the CFSE labeling especially about the stimulation (PHA-stimulation vs antigen, cytokines stimulation). Before we continue, let us specify the experimental procedure for collecting these data.

Heparinized blood samples were collected from two healthy human volunteers at the New York Blood Center. Peripheral blood mononuclear cells were isolated using standard Ficoll/Hypaque density gradient centrifugation and were frozen at -150° Celsius until use. Human subject study

regulations and guidelines were strictly followed during the studies. Regarding the procedure of labeling, it is similar to the one described earlier. To be able to justify that the proliferation is not due to chance but only to stimulation, Liu et al. used a famous technique which consists to divide the sample into two groups, the treatment group and the control group. The control group serves as the baseline measure for the experiment performed on the treatment group. The work was done both for CD4 and CD8 and the distinction was made by looking at the markers present on the surface of cells. By using essentially two statistical software notably **ModFit** and **CellQuest** to record and analyze the results. As a result, the authors showed that cell proliferation can be determined using statistical analysis of CFSE labeling data.

From the above discussion, we can conclude that the available data is trustworthy and completely describe cells proliferation up to the 6th generation count from 0th generation. Therefore, we will use this data to estimate the "best" parameters of the model by fitting the model to this data. The only question here is the goodness of the fitness, given that we have twenty-one parameters to be estimated with forty-nine observations.

5.3 Maximum Likelihood for the available data

5.3.1 Multivariate Likelihood function

Let us suppose that we have T observations (times points in which the observations are made), and we have $h+1$ random variables $X_0, X_1, \cdots, X_h$ that represent the number of cells of each generation as function of time. We model the cell division process using the multivariate random variable

$$X = (X_0, X_1, \cdots, X_h), \qquad X_i \in \{n^i_{k_i}\}_{1\le k_i \le T}. \tag{5.3.1}$$

Each random variable follows a negative binomial distribution, which we parameterize with Poisson-gamma mixture with parameter $(m^i_{k_i}, b^i_{k_i})$, and the corresponding probability is giving by

$$P(X_i = n^i_{k_i}) = \frac{\Gamma(n^i_{k_i} + 1/b^i_{k_i})}{n^i_{k_i}!\Gamma(1/b^i_{k_i})}(m^i_{k_i}b^i_{k_i})^{n^i_{k_i}}(1 + m^i_{k_i}b^i_{k_i})^{-n^i_{k_i}+1/b^i_{k_i}}. \tag{5.3.2}$$

We further suppose that the $h+1$ random variable are pairwise independent, and we introduce the follow notations:

$$n_k = (n^0_{k_0}, n^1_{k_1}, \cdots, n^h_{k_h}),$$

it follows that the join probability is giving as follow

$$\begin{aligned} P(X = n_k) &= P(X_0 = n^0_{k_0}, X_1 = n^1_{k_1}, \cdots, X_h = n^h_{k_h}) \\ &= \prod_{i=0}^{h} P(X_i = n^i_{k_i}) \\ &= \prod_{i=0}^{h} \frac{\Gamma(n^i_{k_i} + 1/b^i_{k_i})}{n^i_{k_i}!\Gamma(1/b^i_{k_i})}(m^i_{k_i}b^i_{k_i})^{n^i_{k_i}}(1 + m^i_{k_i}b^i_{k_i})^{-n^i_{k_i}+1/b^i_{k_i}}. \end{aligned} \tag{5.3.3}$$

It is important to notice that the index i on k_i just specifies which counter we are using and eases the generalization. Let us set up the scene now for the Likelihood function.

5.3.2 likelihood for the data

Let

$$\Omega = \{1, 2, \cdots, T\}^{h+1}, \quad k = \{k_0, k_1, \cdots, k_h\} \quad \text{and} \quad \theta = (\alpha_0, \cdots, \alpha_h, \delta_0, \cdots, \delta_h, \pi_0, \cdots, \pi_h)$$

where each k_i takes it values in $\{1, 2, \cdots T\}$. The likelihood function is given by

$$\begin{aligned} L(\theta, X) &= \prod_{k\in\Omega} P(X|\theta) \\ &= \prod_{k\in\Omega}\left(\prod_{i=0}^{h} \frac{\Gamma(n^i_{k_i} + 1/b^i_{k_i})}{n^i_{k_i}!\Gamma(1/b^i_{k_i})}(m^i_{k_i}b^i_{k_i})^{n^i_{k_i}}(1 + m^i_{k_i}b^i_{k_i})^{-n^i_{k_i}+1/b^i_{k_i}}\right). \end{aligned} \tag{5.3.4}$$

The log-likelihood function is given by

$$
\begin{aligned}
l(\theta, X) \;\;=\;\; & \sum_{k\in\Omega}\sum_{i=0}^{h}\Big[\ln(\Gamma(n^i_{k_i}+1/b^i_{k_i})) - \ln(n^i_{k_i}!) - \ln(\Gamma(1/b^i_{k_i})) + n^i_{k_i}(\ln m^i_{k_i} + \ln b^i_{k_i}) \\
& +(-n^i_{k_i}+1/b^i_{k_i})\ln(1+m^i_{k_i}b^i_{k_i})\Big] \qquad (5.3.5)
\end{aligned}
$$

Remark 3. $m^i_{k_i}$ and $b^i_{k_i}$ depend on θ.

In our case, $h = 6$ corresponding to the generation up to which we have data. $T = 7$, corresponding to the time points of observations. The $n^i_{k_i}$ correspond to an observation for the ith generation at the time point k_i.

It is also important to notice that with the available data gives cells' dynamic without distinguishes quiescent to activated cells, therefore gives the sum of two types, it follows that in our case, the means and dispersion parameters as function of time (t=x) are given by:

$$
m^i_{k_i} = \left[\mathbf{E}\;(n^{(Q)}_i + n^{(A)}_i)\right]\Big|_{x=k_i} = \left[\mathbf{E}\;(n^{(Q)}_i) + \mathbf{E}\;(n^{(A)}_i)\right]\Big|_{x=k_i}, \qquad (5.3.6)
$$

$$
b^i_{k_i} = \left[\frac{\mathbf{Var}\;(n^{(Q)}_i) + \mathbf{Var}\;(n^{(A)}_i) + 2\times\mathbf{Cov}\;(n^{(Q)}_i, n^{(A)}_i) - m^i_{k_i}}{\left(m^i_{k_i}\right)^2}\right]\Bigg|_{x=k_i}. \qquad (5.3.7)
$$

With this set up, it remains to define the space where there parameters lie in. In the literature (see [41]) the condition to have the existence of the maximum estimator is that

$$
b^i_{k_i} > -\frac{1}{\max(n^i_{k_i})}, \quad 0 \le i \le h. \qquad (5.3.8)
$$

Therefore the algorithm should take that into account. The next problem arising is the starting point for our parameters (initialization). for that, once we clearly know the range of each parameters, we can draw the initial parameters from a random uniform distribution.

Now that everything is set up, it remains to address the question of the lack of data for parameter estimation. To sort out this issue based of the remark by Gett and Hodgkin [16] on *Three-parameters model of T-cell activation* where the authors noted that the time taken to complete the first division is different from the times for subsequent divisions which seem to be constant.

Therefore, we will make the following assumptions:

$$\alpha_i = \begin{cases} \alpha_0 & \text{if } i = 0 \\ \alpha & \text{otherwise,} \end{cases} \qquad \delta_i = \begin{cases} \delta_0 & \text{if } i = 0 \\ \delta & \text{otherwise} \end{cases} \qquad \text{and} \quad \pi_i = \begin{cases} \pi_0 & \text{if } i = 0 \\ \pi & \text{otherwise.} \end{cases} \tag{5.3.9}$$

which reduce the 21 parameters to only 6.

We have constructed the theoretical and practical framework for estimating the parameters of T cell proliferation. It remains to implement the MLE to estimate the parameters.This objective will be pursued in the continuation of our work

6. Conclusion

The CFSE labeling is presented as one of the best experimental techniques used to quantify T cells' proliferation. However, the experimental data produced does not provide direct information about the underlying parameters. To be able to extract new information and estimate the parameters, we revisited the experimental protocol for monitoring the dynamics of T lymphocytes using the CFSE dye. The works by Quah et al. [14] and Quah and Parish [3] allowed us to better understand the cell biology and the assumptions to be taking into account for modeling the proliferation process. Thereafter, we revisited some mathematical models that have served to interpret the CFSE labeling experiment. Based on the assumptions extract from CFSE labeling and knowledge from cell biology, we reviewed the earlier model by Juwara [23] that describe the T-cell proliferation as a discrete Markov jump process. Using the Chapman Kolmogrov equation, we ended with stochastic master equation (SME) describing the studied process.

The obtained model is not directly solvable. As solution, we used the probability generating function and the method of characteristic to solve it implicitly. The explicit solution can be obtained recursively, which in practice is difficult. Nevertheless, we extracted some information, that is to say the means of cell counts, the variances of cell counts and the covariances of cell counts in the case where activation rate is no time dependent. By observing the process we concluded that it could be described by the negative binomial with the parameters extracted earlier. We ended by constructing the framework for fitting the obtained model to data and estimate the parameters of the model.Those are the activation rates, the division rates and the death rates.

As we mentioned earlier, the motivation of this work was first to build a flexible model that could include as much information as possible about T cell proliferation dynamics, especially the recent insight by Heinzel et al. [18] that it would be the molecule of Myc that governs the proliferation process via activation process. Hence, the activation parameter is, on the one hand, time-dependent and on the other hand threshold-dependent. The second motivation was that of constructing a framework for parameters estimation using the maximum likelihood approach. These goals had been both largely achieved. The next step will be to first adjust the model to the data in order to estimate the parameters in the case where the activation parameter is constant over the time. We will then calculate the means, variances, and covariances in the case where the activation parameter is time-dependent. We will finally use this information to fit the model to the data in order to estimate the parameters of the model. We intend to achieve these new goals in our future work in PhD.

7. Appendix

7.1 CFSE experimental Protocol

Originally, the CFSE is a fluorescent dye was developed to track lymphocyte migration and positioning within animal bodies for many months [13] . Later [4] showed that the dye can also be used to monitor lymphocyte proliferation, both in vitro and in vivo in view of the gradual reduction of half the intensity of the dye in the daughter cells after each division. Moreover, when used optimally, it maintains cell viability and allows cell division to be monitored up to the 8th generation. An even more interesting feature of this technique is that of division-dependence of changes in the expression of cell surface markers, so generations can be easily identified by flow cytometry [5]. In addition, subsets of lymphocytes in complex mixtures of cells can be easily quantified using the appropriate cell surface[4]. Here we give a description of the experimental procedure inspired by the works done by Quah et al. [14] and Quah and Parish [3].

7.1.1 Materials

To perform the CFSE labeling, there are a list of materials required:

- Carboxyfluorescein diacetate succinimidyl ester (CFDA, SE) (MW 557.47 g/mole) which will be converted into CFSE after the release of acetate group
- Dimethyl sulfoxide (DMSO)
- Fresh T cell from human or from mouse
- RPMI 1640 medium
- Fetal calf serum (FCS)
- 10-15 mL of canonical tubes
- Phosphate buffered saline (PBS)
- Aluminium foil
- vortex

- Centrifuge
- Flow cytometer.

7.1.2 Reagent setup

- Dissolve 25mg of CFDA, SE power in 8,96 mL of DMSO to obtain final solution of 5mM.(be careful because the CFSE reacts rapidly with the solution DMSO. It is advised to do this mixture just before the cell labeling.
- Freshly isolated T cell from spleen or lymph nodes of human or mice and resuspend in 1 mL of culture medium (RMPI) with 5% heat-inactivated (HI) fetal calf serum (FCS), with a cell concentration of between 0.5×10^6- 10×10^7 mL.
- For cell purification reagent, one uses negative selection procedure by using magnetic bead technologies, positive or negative selection procedures using magnetic microbeads or cell sorting using Immunofluorescence flow cytomery.
- Use the appropriate stimuli to encourage the proliferation, for instance phytohemagglutinin is indicated for human T cell and ovalbumin for mouse.
- To distinguish the dead cells from viable cells, propidiumiodide or 7-aminoactinomycin D is indicated.

7.1.3 CFSE labeling of T cells

For the Labeling, we describe the routine method given by [3]:

- Thoroughly resuspend cells in the 1 mL volume of culture medium and place carefully in the bottom of a fresh (non-wetted) 10 mL conical tube.
- Hold the tube horizontally to prevent premature mixing of suspended cell and CFDA, SE solution.
- Add 110μL of PBS to the dry part at the top tube and carefully to avoid contact with cell solution.

- Resuspend $1.1\ \mu L$ of the 5 mM stock of CFDA, SE in the 110 μL of PBS.
- Quickly mixing to obtain uniform mixture
- Cover the tube with aluminuim foil to protect it from the light and incubate cells for five minutes at room temperature.
- Wash cells by diluting in 10 volumes of 20 degree Celsuis of PBS containing 5% of HI FCS, sedimenting by centrifugation at $300\times$ g for 5 min at 20 degree Celsuis and discarding the supernatant. Repeat washing twice more.
- Proceed to the counting by using cytometry flow.

The molecular events that occurs during the CFSE labeling of T cells can be summarize by the figure (7.1).

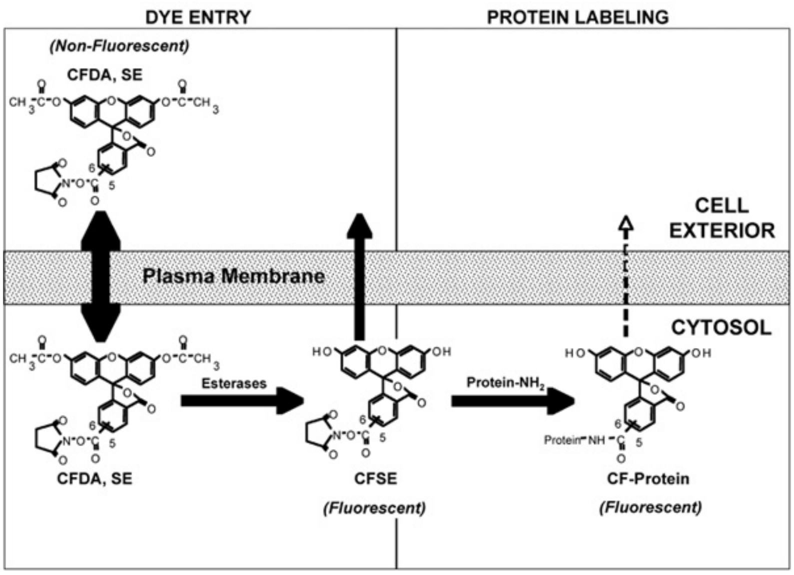

Figure 7.1: A representation of the various molecular events that occur during the labeling of cells with carboxyfluorescein diacetate succinimidyl ester (CFDA, SE) Image from [3]

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